

**MAR ATHANASIOS COLLEGE FOR ADVANCED STUDIES, TIRUVALLA
(MACFAST)**

(Affiliated to Mahatma Gandhi University, Kottayam)



MACFAST (AUTONOMOUS)

MCACP 205 OBJECT ORIENTED LAB (JAVA LAB)

Department of Computer Applications

Mar Athanasios College for Advanced Studies, Tiruvalla

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(MACFAST)

(Affiliated to Mahatma Gandhi University, Kottayam)



MACFAST (AUTONOMOUS)

MCACP 205 OBJECT ORIENTED LAB (JAVA LAB)

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Autonomous).*

Faculty in Charge

Head of the Department

Examiners

1.....

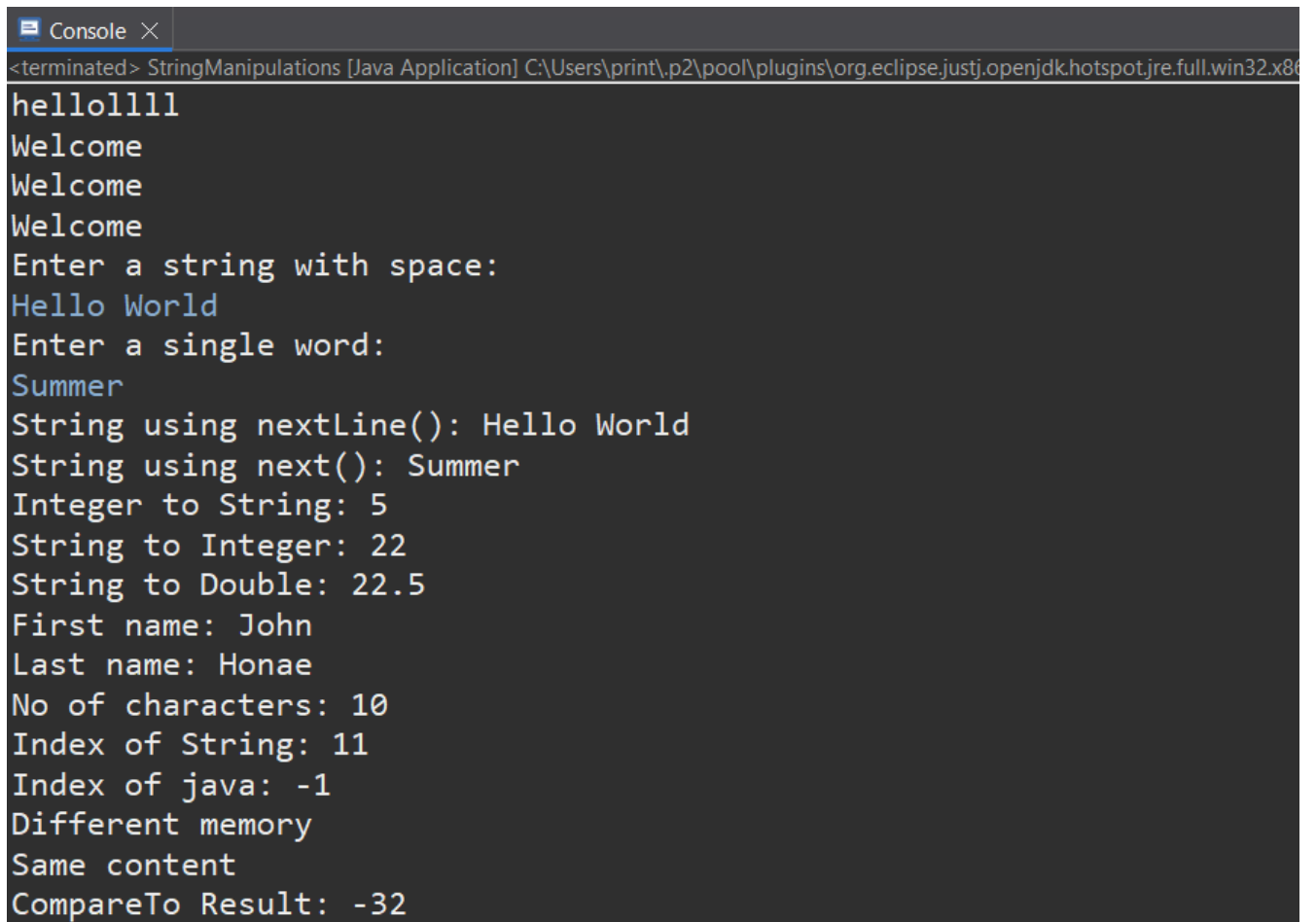
2.....

INDEX

SL.NO	DATE	NAME OF PROGRAM	PAGE NO
1	18/02/2026	String Manipulation	6
2	18/02/2026	Array	12
3	20/02/2026	Array 2D User Input – Matrix formation	14
4	20/02/2026	Biggest Of 3 Numbers	16
5	20/02/2026	Binary Search	18
6	25/02/2026	Demonstrate Classes & Objects Using Employee Class	22
7	25/02/2026	Demonstrate Interface Payment	26
8	27/02/2026	Demonstrate Inheritance Using Person & Faculty classes	30
9	27/02/2026	Demonstrate Abstract Class	36
10	04/03/2026	Demonstrate Exception Handling	42
11	04/03/2026	Demonstrate Multiple Exception Handling	44
12	04/03/2026	Demonstrate User-Defined Exception Handling	46
13	05/03/2026	Demonstrate Thread	48
14	05/03/2026	Demonstrate Event Handling Using Action Listener	50
15	11/03/2026	Demonstrate File Creation	54

16	11/03/2026	Demonstrate File Writing	56
17	11/03/2026	Demonstrate File Reading	58
18	11/03/2026	Demonstrate File Appending	60
19	12/03/2026	Demonstrate Package Creation	62
20	12/03/2026	Demonstrate Layout Manager	66
21	18/03/2026	Demonstrate GUI Calculator	72
22	18/03/2026	Demonstrate Menu Creation	82
23	19/05/2026	Demonstrate Applet Using Human Face	86
24	19/05/2026	Demonstrate Database Connectivity with MySQL	90
25	26/05/2026	Demonstrate AWT Components Using Registration Form	94
26	26/05/2026	Demonstrate Socket Programming	100

OUTPUT



The screenshot shows a console window titled "Console" with a close button. The window contains the following text:

```
<terminated> StringManipulations [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64.jdk\bin\java.exe
hello1111
Welcome
Welcome
Welcome
Enter a string with space:
Hello World
Enter a single word:
Summer
String using nextLine(): Hello World
String using next(): Summer
Integer to String: 5
String to Integer: 22
String to Double: 22.5
First name: John
Last name: Honae
No of characters: 10
Index of String: 11
Index of java: -1
Different memory
Same content
CompareTo Result: -32
```

PROGRAM – 1

AIM : Write a Java program to perform various string manipulation operations.

SOURCE CODE:

```
package labprgms;
import java.util.Scanner;
public class StringManipulations {
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    String str1 = new String("hellollll");
    System.out.println(str1);
    String str2 = "Welcome";
    System.out.println(str2);
    char[] data = { 'W', 'e', 'l', 'c', 'o', 'm', 'e' };
    // prints array as string
    System.out.println(data);
    String str3 = new String(data);
    System.out.println(str3);
    System.out.println("Enter a string with space:");
    String str4 = sc.nextLine();
    System.out.println("Enter a single word:");
    String str5 = sc.next();
    System.out.println("String using nextLine(): " + str4);
    System.out.println("String using next(): " + str5);
    // Convert int to String
    int num = 5;
    String str6 = String.valueOf(num);
    System.out.println("Integer to String: " + str6);
    // Convert String to int
    String str7 = "22";
    int intValue = Integer.parseInt(str7);
    System.out.println("String to Integer: " + intValue);
}
```



```

// Convert String to double
String str8 = "22.5";
double doubleValue = Double.parseDouble(str8);
System.out.println("String to Double: " + doubleValue);
// substring()
String fullName = "John Honae";
// substring(startIndex, endIndex)
String firstName = fullName.substring(0, 4);
String lastName = fullName.substring(5);
System.out.println("First name: " + firstName);
System.out.println("Last name: " + lastName);
// length()
System.out.println("No of characters: " + fullName.length());
// indexOf()
String str9 = "Welcome to String manipulation and String";
int index1 = str9.indexOf("String");
int index2 = str9.indexOf("java");
System.out.println("Index of String: " + index1);
System.out.println("Index of java: " + index2);
// String Comparison
String str10 = new String("Hello");
String str11 = new String("Hello");
String str12 = "hello";
if (str10 == str11) {
    System.out.println("Same memory");
} else {
    System.out.println("Different memory");
}
// Compare content
if (str10.equals(str11)) {
    System.out.println("Same content");
} else {
    System.out.println("Different content");
}

```



```
// compareTo()
System.out.println("CompareTo Result: " + str11.compareTo(str12));
    sc.close();
}
}
```

OUTPUT

```
Console X
<terminated> Array_user_input [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win
Enter 5 values
6
19
4
17
21
Array Contents
6      19      4      17      21
```

PROGRAM – 2

AIM : Write a Java program to create an array & display elements of Array.

SOURCE CODE:

```
package labprgms;
import java.util.Scanner;
public class Array_user_input {
    public static void main(String[] args) {
        int [] numbers = new int[5];
        Scanner input = new Scanner(System.in);
        System.out.println("Enter 5 values");
        for(int i = 0;i<numbers.length;i++) {
            numbers[i] = input.nextInt();
        }
        System.out.println("Array Contents");
        for(int value:numbers) {
            System.out.print(value+"\t");
        }
    }
}
```

OUTPUT

```
Console X
<terminated> Array2D [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_2
Enter 6 values:
21
33
18

07
10
11
|
----- Matrix -----
21      33      18
7       10      11
```

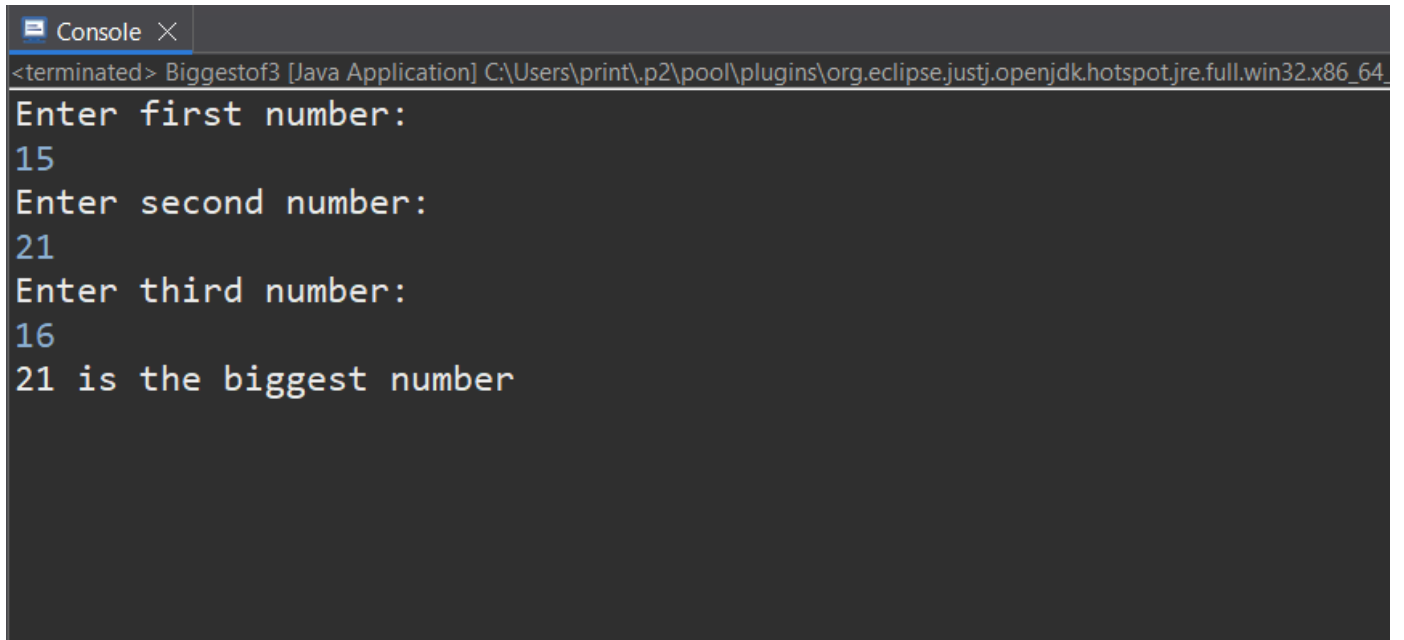
PROGRAM – 3

AIM : Write a Java program to display the elements of a 2D array.

SOURCE CODE:

```
package labprgms;
import java.util.*;
public class Array2D {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int [][] matrix = new int [2][3];
        System.out.println("Enter 6 values:");
        for(int row =0;row<matrix.length;row++)
        {
            for(int col =0;col<matrix[row].length;col++)
            {
                matrix[row][col]= input.nextInt();
            }
            System.out.println();
        }
        System.out.println("----- Matrix -----");
        for(int [] row: matrix) {
            for(int value:row) {
                System.out.print(value+"\t");
            }
            System.out.println();
        }
    }
}
```

OUTPUT



```
Console X
<terminated> Biggestof3 [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64
Enter first number:
15
Enter second number:
21
Enter third number:
16
21 is the biggest number
```

PROGRAM – 4

AIM : Write a Java program to find the largest among three numbers.

SOURCE CODE:

```
package labprgms;
import java.util.Scanner;
public class Biggestof3 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int a, b, c;
        System.out.println("Enter first number:");
        a = sc.nextInt();
        System.out.println("Enter second number:");
        b = sc.nextInt();
        System.out.println("Enter third number:");
        c = sc.nextInt();

        if (a > b && a > c) {
            System.out.println(a + " is the biggest number");
        }
        else if (b > a && b > c) {
            System.out.println(b + " is the biggest number");
        }
        else {
            System.out.println(c + " is the biggest number");
        }
        sc.close();
    }
}
```

OUTPUT

```
Console X
<terminated> binarysearch [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.j
Enter 5 elements:
19
3
56
13
47
sorted :
3      13      19      47      56
enter the element to search:
47
47 found at 4
|
```

PROGRAM – 5

AIM : Write a Java program to search for an element in a sorted array using the Binary Search.

SOURCE CODE:

```
package labprgms;
import java.util.Scanner;
import java.util.Arrays;
public class BinarySearch {
    public static int search(int [] num, int search)
    {
        int low=0;
        int high=num.length - 1;
        while(low < high)
        {
            int mid = (low + high )/2;

            if(num[mid] == search)
            {
                return mid + 1;
            }
            else if( num[mid] < search)
            {
                low = mid + 1;
            }
            else {
                high = mid - 1;
            }
        }
        return -1;
    }
}
```

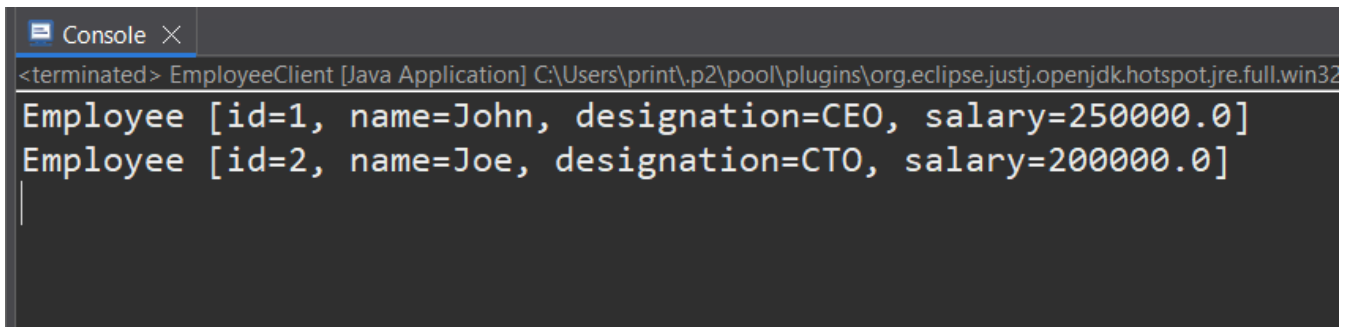


```

public static void main(String[] args) {
    Scanner input = new Scanner(System.in);
    System.out.println("Enter 5 elements:");
    int [] nos = new int[5];
    for(int i = 0;i<nos.length;i++)
    {
        nos[i] = input.nextInt();
    }
    Arrays.sort(nos);
    System.out.println("sorted :");
    for(int value:nos)
    {
        System.out.print(value+"\t");
    }
    System.out.println();
    System.out.println("enter the element to search:");
    int s = input.nextInt();
    input.close();
    int loc = search(nos,s);
    if(loc == -1)
    {
        System.out.println("Not found");
    }
    else {
        System.out.println(s +" found at "+loc);
    }
}
}

```

OUTPUT



```
<terminated> EmployeeClient [Java Application] C:\Users\print\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32
Employee [id=1, name=John, designation=CEO, salary=250000.0]
Employee [id=2, name=Joe, designation=CTO, salary=200000.0]
```

PROGRAM – 6

AIM : Write a Java program to implement class and object by using Employee & EmployeeClient class.

SOURCE CODE:

Employee.java

```
package labprgms;

public class Employee {
    private int id;
    private String name;
    private String designation;
    private double salary;
    public Employee(){
        // default constructor
        name = "";
        id = 0;
        designation = "";
        salary = 0.0;
    }
    public Employee(int id, String n,String desig, double salary) {
        // Parameterized constructor
        this.id = id;
        name = n;
        designation = desig;
        this.salary = salary;
    }
    public int getId() {
        return id;
    }
    public void setId(int id) {
        this.id = id;
    }
    public String getName() {
        return name;
    }
}
```



```

    }
    public void setName(String name) {
        this.name = name;}
    public String getDesignation() {
        return designation;}
    public void setDesignation(String designation) {
        this.designation = designation;}
    public double getSalary() {
        return salary;}
    public void setSalary(double salary) {
        this.salary = salary;}
    public String toString() {
        return "Employee [id=" + id + ", name=" + name
            + ", designation=" + designation + ", salary=" + salary + "]);
    }
}

```

EmployeeClient.java

```

package labprgms;
import java.util.Scanner;
public class EmployeeClient {
    public static void main(String[] args) {
        Employee e1 = new Employee();
        e1.setId(1);
        e1.setName("John");
        e1.setDesignation("CEO");
        e1.setSalary(250000);
        Employee e2 = new Employee(2,"Joe", "CTO", 200000);
        System.out.println(e1);
        System.out.println(e2);
    }
}

```

OUTPUT

```
Console X
<terminated> TestPayment [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.j
Enter the amount
500
Service Charges for 500.0 in Credit Card : 50.0
Service Charges for 500.0 in NetBanking : 10.0
Service Charges for 500.0 in Pay Pal : 5.0
```

PROGRAM – 7

AIM : Write a Java program to demonstrate the implementation of an interface.

SOURCE CODE:

Payment.java

```
package labprgms;

public interface Payment {

    public double servicePay(double amount);

}
```

NetBanking.java

```
package labprgms;

public class NetBanking implements Payment{

    public double servicePay(double amount) {

        return amount * 0.02;

    }

}
```

PayPal.java

```
package labprgms;

public class PayPal implements Payment{

    public double servicePay(double amount) {

        return amount * 0.01;

    }

}
```


CreditCard.java

```
package labprgms;

public class CreditCard implements Payment {
    private double amount;
    public CreditCard() {
        amount = 0;
    }
    public CreditCard(double amt) {
        amount = amt;
    }
    public double servicePay(double amount) {
        return amount * 0.1;
    }
}
```

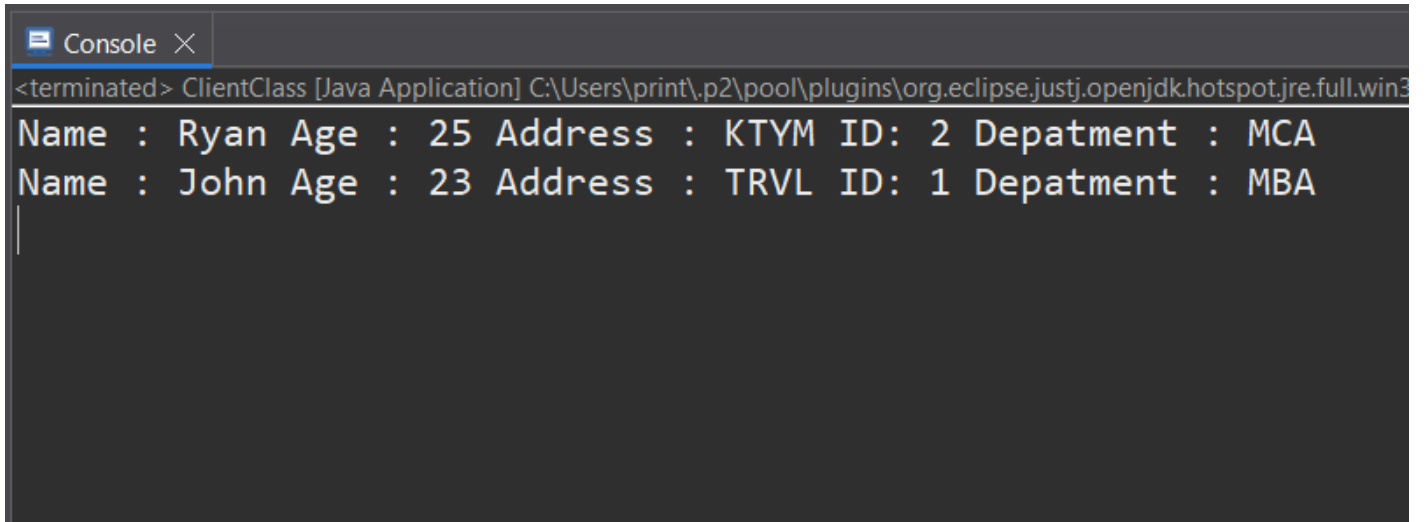
TestPayment.java

```
package labprgms;

import java.util.Scanner;

public class TestPayment {
    public static void main(String[] args) {
        CreditCard c1 = new CreditCard();
        NetBanking n1 = new NetBanking();
        PayPal p1 = new PayPal();
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the amount");
        double payment = sc.nextDouble();
        System.out.println("Service Charges for "+payment+" in Credit Card : "+c1.servicePay(payment));
        System.out.println("Service Charges for "+payment+" in NetBanking : "+n1.servicePay(payment));
        System.out.println("Service Charges for "+payment+" in Pay Pal : "+p1.servicePay(payment));
    }
}
```

OUTPUT

A screenshot of an Eclipse IDE console window. The title bar shows a 'Console' tab with a close button. The console text is as follows:

```
<terminated> ClientClass [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win3  
Name : Ryan Age : 25 Address : KTYM ID: 2 Depatment : MCA  
Name : John Age : 23 Address : TRVL ID: 1 Depatment : MBA  
|
```

PROGRAM – 8

AIM: Write a Java program to demonstrate inheritance.

SOURCE CODE:

Person.java

```
package labprgms;

public class Person {
    private String name;
    private int age;
    private String address;
    public Person(){
        name = "UNK";
        age = 0;
        address = "UNK";
    }
    public Person(String n,int a, String add) {
        name = n;
        age = a;
        address = add;
    }
    public void setName(String n) {
        name = n;
    }
    public void setAge(int a) {
        age = a;
    }
    public void setAddress(String a) {
        address = a;
    }
    public String getName() {
        return name;
    }
}
```



```

    public int getAge() {
        return age;
    }
    public String getAdd() {
        return address;
    }
    public String toString() {
        return "Name : "+name + " Age : " +age + " Address : " +address;
    }
}

```

Faculty.java

```

package labprgms;

public class Faculty extends Person{
    private int facId;
    private String department;
    public Faculty() {
        super();
        facId = 0;
        department = "UNK";
    }
    public Faculty(String n,int a, String add,int id, String dept) {
        super(n,a,add);// calls the parent class constructor
        facId = id;
        department = dept;
    }
    public void setFacID(int id) {
        facId = id;
    }
    public void setDept(String dept) {
        department = dept;
    }
}

```



```

    public int getFacID() {
        return facId;
    }
    public String getDepartment() {
        return department;
    }
    public String toString() {
        return super.toString()+ " ID: " + facId + " Department : "+department;
    }
}

```

ClientClass.java

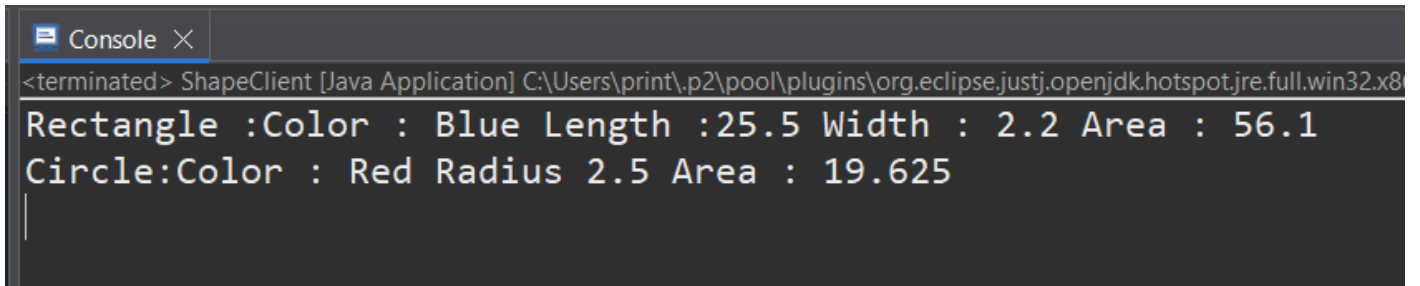
```

package labprgms;

public class ClientClass {
    public static void main(String[] args) {
        Faculty f1 = new Faculty();
        Faculty f2 = new Faculty("John",23,"TRVL", 1, "MBA");
        f1.setName("Ryan");
        f1.setAge(25);
        f1.setAddress("KTYM");
        f1.setFacID(2);
        f1.setDept("MCA");
        System.out.println(f1);
        System.out.println(f2);
    }
}

```

OUTPUT



```
Console X
<terminated> ShapeClient [Java Application] C:\Users\print\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64.jdk\bin\java.exe
Rectangle :Color : Blue Length :25.5 Width : 2.2 Area : 56.1
Circle:Color : Red Radius 2.5 Area : 19.625
```

PROGRAM – 9

AIM : Write a Java program to demonstrate the concept of abstraction using an abstract class and its derived classes.

SOURCE CODE:

Shape.java

```
package labprgms;

public abstract class Shape {
    private String color;
    public abstract double area(); // abstract methods
    public Shape(String color) {
        this.color = color;
    }
    public void setColor(String color) {
        this.color = color;
    }
    public String getColor() {
        return color;
    }
    public String toString() {
        return "Color : "+color;
    }
}
```

Rectangle.java

```
package labprgms;

public class Rectangle extends Shape{
    private double length;
    private double width;
    public Rectangle(String color, double length, double width) {
        super(color);
        this.length = length;
        this.width = width;
    }
}
```



```

    }
    public double getLength() {
        return length;
    }
    public void setLength(double length) {
        this.length = length;
    }
    public double getWidth() {
        return width;
    }
    public void setWidth(double width) {
        this.width = width;
    }
    public double area() {
        return length * width;
    }
    public String toString() {
        return "Rectangle :"+super.toString()+ " Length :"+length+" Width : "+width + " Area :
"+area();
    }
}

```

Circle.java

```

package labprgms;

public class Circle extends Shape{
    private double radius;
    public Circle(String color, double radius) {
        super(color);
        this.radius = radius;
    }
    public void setRadius(double radius) {
        this.radius = radius;
    }
}

```



```

    public double getRadius() {
        return radius;
    }
    public double area() {
        return 3.14 * radius*radius;
    }
    public String toString() {
        return "Circle:"+super.toString()+ " Radius "+radius + " Area : "+area();
    }
}

```

ShapeClient.java

```

package labprgms;

public class ShapeClient {
    public static void main(String[] args) {
        Rectangle r1 = new Rectangle("Blue", 25.5, 2.2);
        System.out.println(r1);
        Circle c1 = new Circle("Red", 2.5);
        System.out.println(c1);
    }
}

```

OUTPUT

```
Console X
<terminated> ExceptionExample1 [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205
Division
Enter the first number
17
Enter the second number
0
Cant divide by 0
Program completed
```

```
Console X
<terminated> ExceptionExample1 [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205
Division
Enter the first number
13
Enter the second number
fged
Wrong input
Program completed
```

```
Console X
<terminated> ExceptionExample1 [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205
Division
Enter the first number
20
Enter the second number
5
Quotient : 4
Program completed
```

PROGRAM – 10

AIM : Write a Java program to demonstrate Arithmetic Exception handling.

SOURCE CODE:

```
package labprgms;
import java.util.InputMismatchException;
import java.util.Scanner;
public class ExceptionExample1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Division");
        try{
            System.out.println("Enter the first number");
            int num1 = sc.nextInt();
            System.out.println("Enter the second number");
            int num2 = sc.nextInt();
            int qt = num1/num2;
            System.out.println("Quotient : "+qt);
        }
        catch(InputMismatchException e) {
            System.out.println("Wrong input");
        }
        catch(ArithmeticException e) {
            System.out.println("Cant divide by 0");
        }
        catch(Exception e) {
            System.out.println("Some error occurred : "+e.getMessage());
        }
        finally {
            System.out.println("Program completed");
        }
    }
}
```

OUTPUT

```
Console X
<terminated> ExceptionExample2 [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64
Enter the index (0,1 or 2) to print the corresponding value
7
Index out of range (0 to 2)
Program completed
```

```
Console X
<terminated> ExceptionExample2 [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64
Enter the index (0,1 or 2) to print the corresponding value
jgh
Wrong input
Program completed
```

```
Console X
<terminated> ExceptionExample2 [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64
Enter the index (0,1 or 2) to print the corresponding value
0
Value : 25
Program completed
```

PROGRAM – 11

AIM : Write a Java program to demonstrate handling multiple exceptions using multiple catch blocks.

SOURCE CODE:

```
package labprgms;
import java.util.InputMismatchException;
import java.util.Scanner;
public class ExceptionExample2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        try {
            int [] marks = {25,26,58};
            System.out.println("Enter the index (0,1 or 2) to print the corresponding value");
            int index = sc.nextInt();
            System.out.println("Value : "+marks[index]);
        }
        catch(InputMismatchException e) {
            System.out.println("Wrong input");
        }
        catch(ArrayIndexOutOfBoundsException e) {
            System.out.println("Index out of range (0 to 2)");
        }
        catch(Exception e) {
            System.out.println("Some error occurred : "+e.getMessage());
        }
        finally {
            System.out.println("Program completed");
        }
    }
}
```

OUTPUT

```
Console X
<terminated> Emailvalidate (1) [Java Application] C:\Users\print\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20210810
Enter Email ID: Macfast2026@gmail.com
Valid Email ID
```

```
Console X
<terminated> Emailvalidate (1) [Java Application] C:\Users\print\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20210810
Enter Email ID: Macfast2026
Invalid Email ID
```

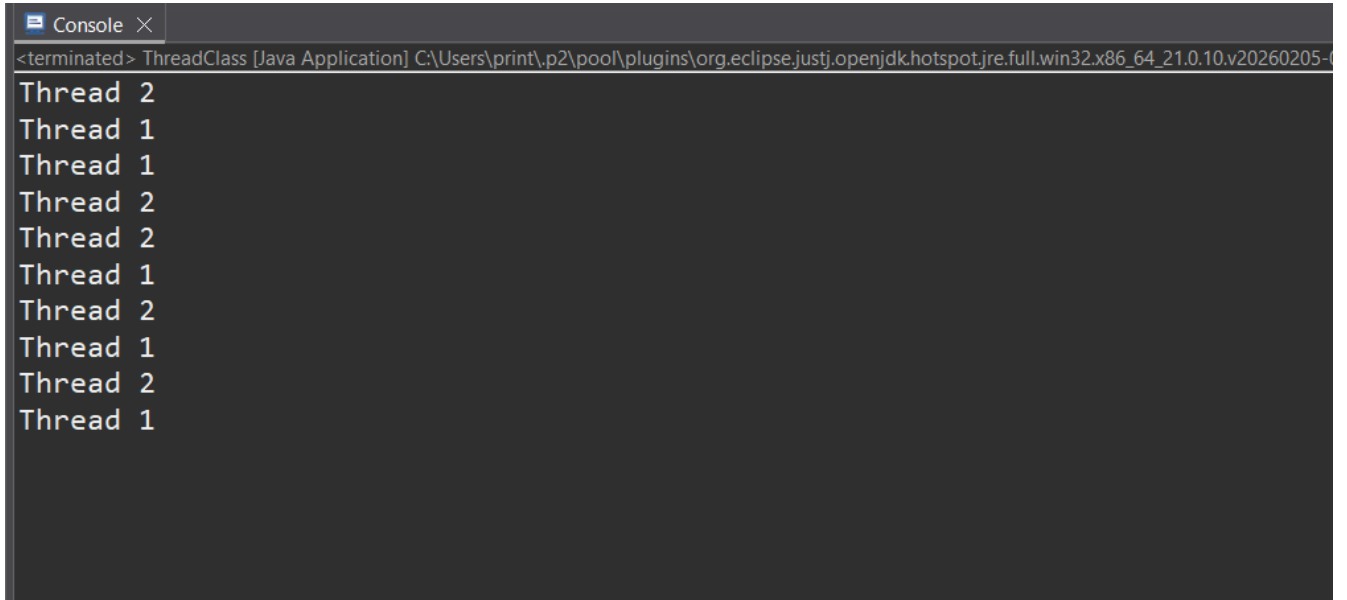
PROGRAM – 12

AIM : Write a Java program to create and handle a user-defined exception.

SOURCE CODE:

```
package labprgms;
import java.util.Scanner;
class InvalidEmailException extends Exception {
    InvalidEmailException(String msg) {
        super(msg);
    }
}
public class Emailvalidate {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Email ID: ");
        String email = sc.nextLine();
        try {
            if (!email.contains("@") || !email.contains(".")) {
                throw new InvalidEmailException("Invalid Email ID");
            }
            System.out.println("Valid Email ID");
        } catch (InvalidEmailException e) {
            System.out.println(e.getMessage());
        }
        sc.close();
    }
}
```

OUTPUT



The screenshot shows an Eclipse IDE console window with a dark background. The title bar at the top reads "Console" with a close button. The console output is as follows:

```
<terminated> ThreadClass [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-0
Thread 2
Thread 1
Thread 1
Thread 2
Thread 2
Thread 1
Thread 2
Thread 1
Thread 2
Thread 1
```

PROGRAM – 13

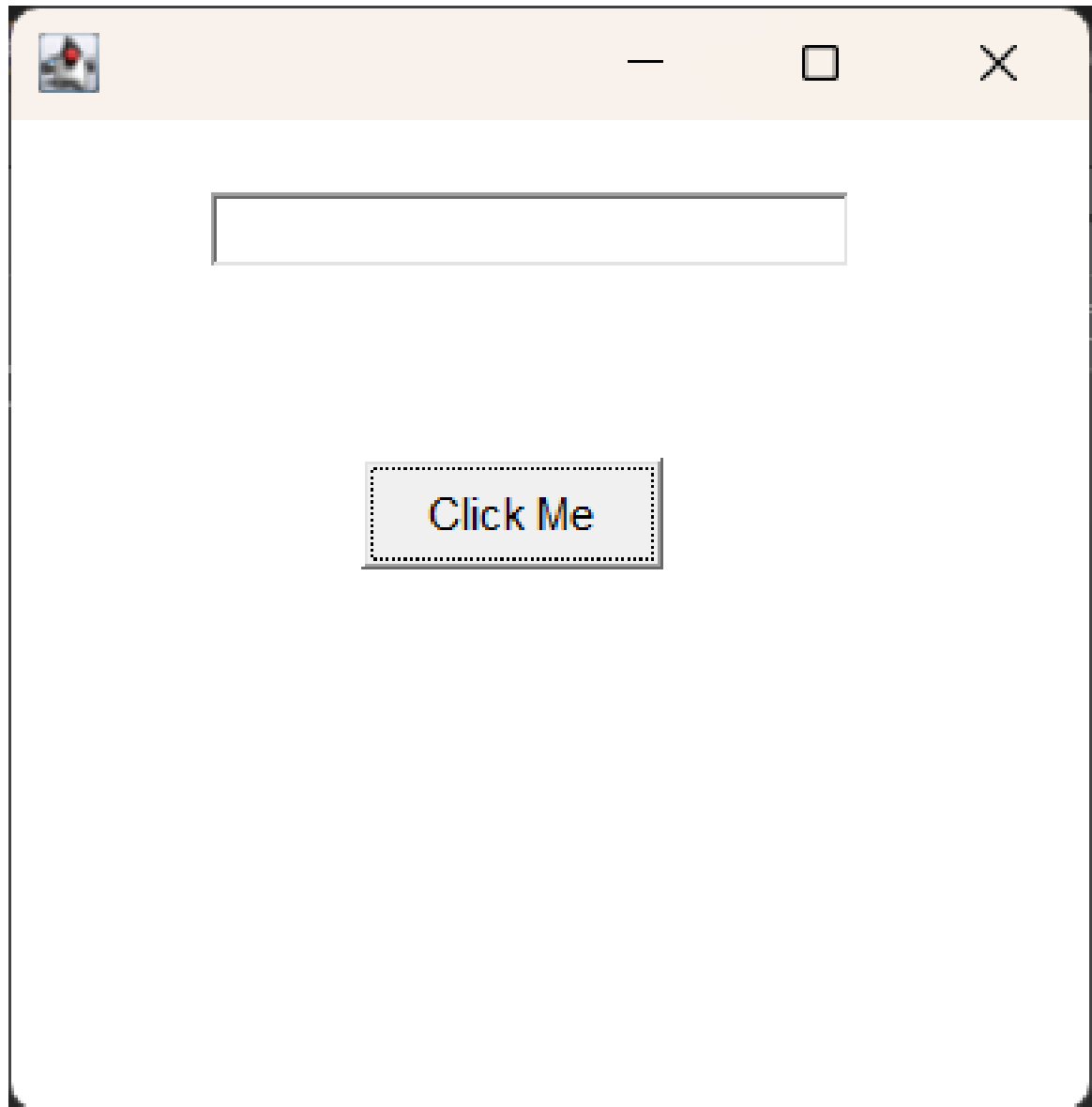
AIM : Write a Java program to demonstrate multithreading by creating and executing threads.

SOURCE CODE:

```
package printpkg;

public class ThreadClass extends Thread {
    private String message;
    public ThreadClass(String message) {
        this.message = message;
    }
    public void run()
    {
        for(int i = 0; i<5; i++)
        {
            System.out.println(message);
            try {
                Thread.sleep(1000);
            }
            catch(InterruptedException e){
                e.printStackTrace();
            }
        }
    }
    public static void main(String[] args) {
        ThreadClass t1 = new ThreadClass("Thread 1");
        ThreadClass t2 = new ThreadClass("Thread 2");
        t1.start();
        t2.start();
    }
}
```

OUTPUT



Before Clicking on **Click Me** Button

PROGRAM – 14

AIM : Write a Java program to demonstrate event handling using ActionListener in a GUI application.

SOURCE CODE:

```
package labprgms;
import java.awt.*;
import java.awt.event.*;
class EventExample extends Frame implements ActionListener
{
    TextField tf;
    EventExample()
    {
        // Create TextField
        tf = new TextField();
        tf.setBounds(60, 50, 170, 20);

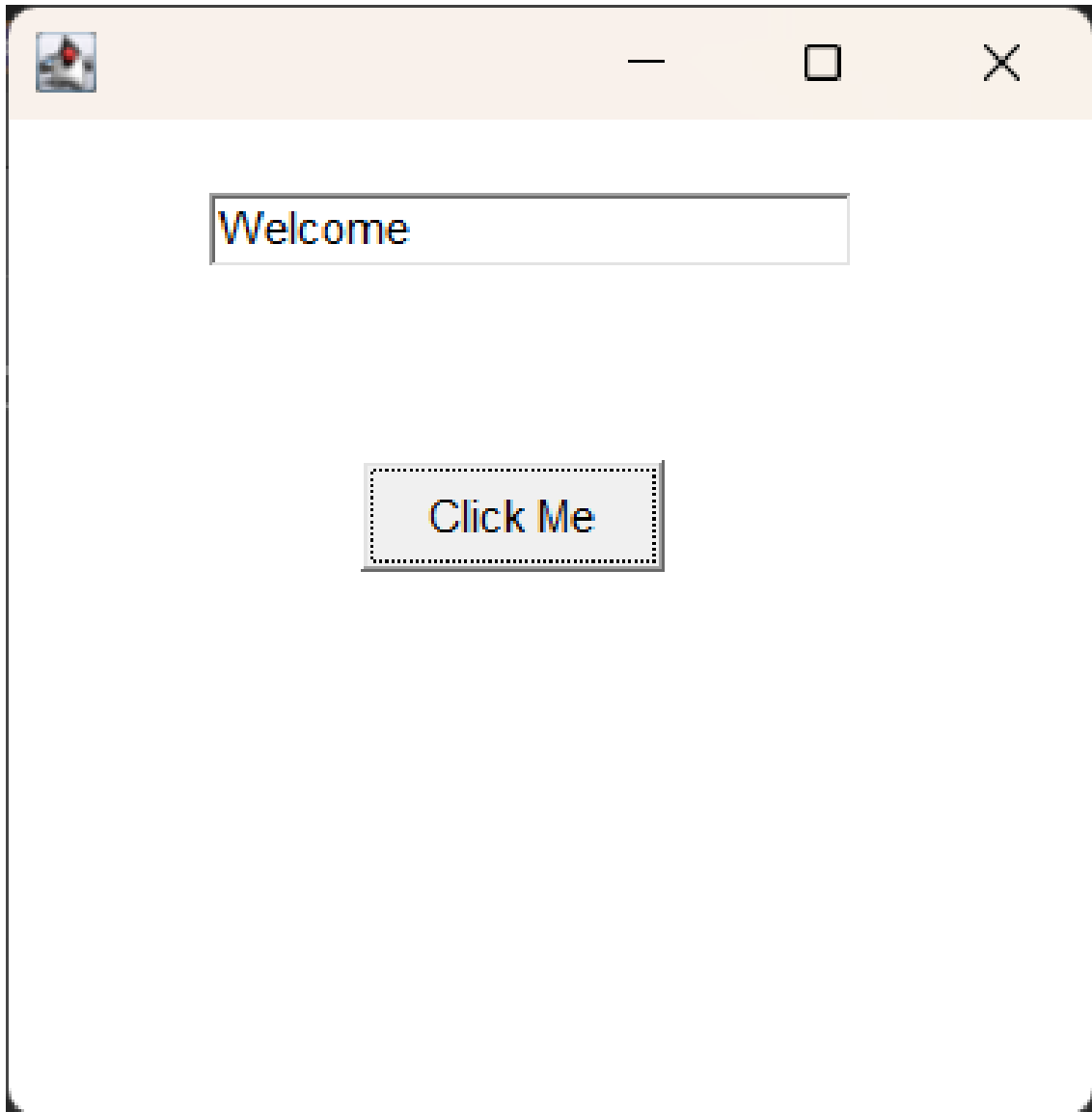
        // Create Button
        Button b = new Button("Click Me");
        b.setBounds(100, 120, 80, 30);

        // Register ActionListener
        b.addActionListener(this);

        // Add components
        add(b);
        add(tf);

        // Frame settings
        setSize(300, 300);
        setLayout(null);
        setVisible(true);
    }
}
```

OUTPUT



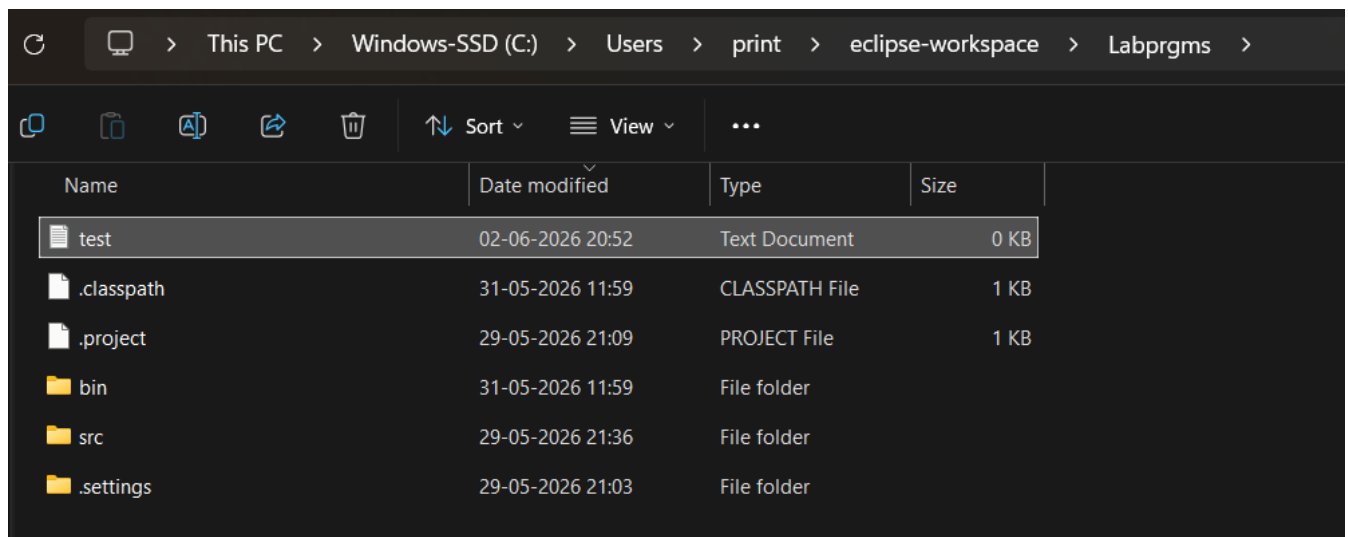
After Clicking on **Click Me** Button

```
// Event Handling Method
public void actionPerformed(ActionEvent e)
{
    tf.setText("Welcome");
}

// Main Method
public static void main(String args[])
{
    new EventExample();
}
}
```

OUTPUT

```
Console X
<terminated> FileCreation (1) [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-0638\jre
File Created at C:\Users\print\eclipse-workspace\Labprgms\test.txt
```



Name	Date modified	Type	Size
test	02-06-2026 20:52	Text Document	0 KB
.classpath	31-05-2026 11:59	CLASSPATH File	1 KB
.project	29-05-2026 21:09	PROJECT File	1 KB
bin	31-05-2026 11:59	File folder	
src	29-05-2026 21:36	File folder	
.settings	29-05-2026 21:03	File folder	

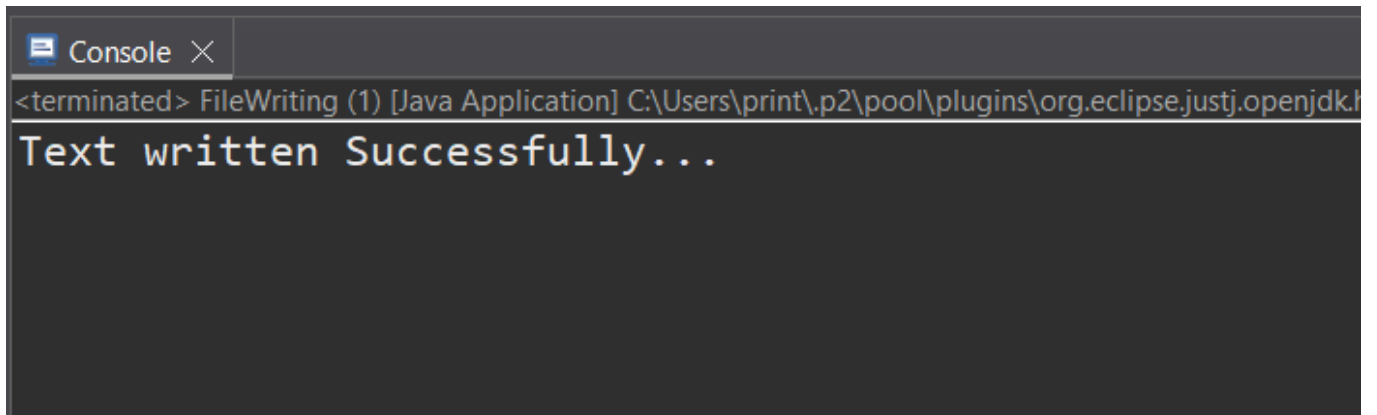
PROGRAM – 15

AIM : Write a Java program to create a file, check its existence, and display an appropriate message.

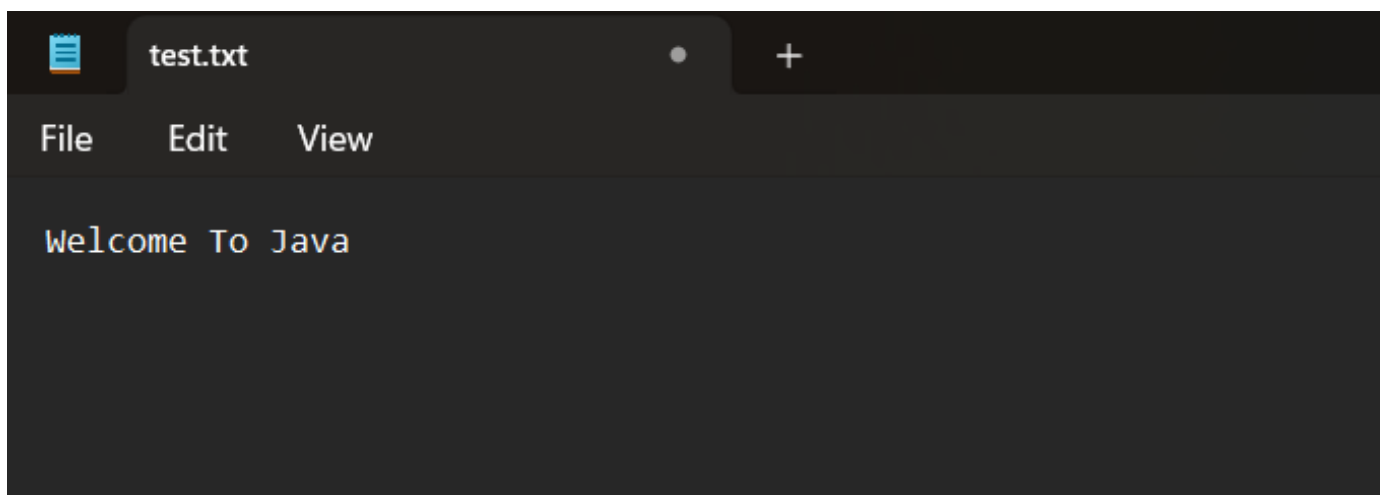
SOURCE CODE:

```
package printpkg;
import java.io.*;
public class FileCreation {
    public static void main(String[] args) {
        // File Creation
        File myFile = new File("test.txt");
        try {
            if(myFile.createNewFile() == true)
            {
                System.out.println(" File Created at "+myFile.getAbsolutePath());
            }
            else {
                System.out.println("Not able to create");
            }
        }
        catch(IOException e){
            e.printStackTrace();
        }
    }
}
```

OUTPUT



The screenshot shows the Eclipse IDE's Console window. The title bar reads 'Console' with a close button. The console output displays the message: `<terminated> FileWriting (1) [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.h` followed by `Text written Successfully...` on the next line.



The screenshot shows the Eclipse IDE's editor window. The title bar indicates the file is 'test.txt'. The menu bar includes 'File', 'Edit', and 'View'. The editor area contains the text: `Welcome To Java`.

PROGRAM – 16

AIM : Write a Java program to create a file and write data into it using file handling classes.

SOURCE CODE:

```
package labprgms;

import java.io.*;

public class FileWriting {

    public static void main(String[] args) {

        try {

            // write() --> Overwrite the entire file/ write into the file.

            FileWriter writer = new FileWriter("test.txt");

            writer.write("Welcome To Java");

            writer.close();

            System.out.println("Text written Successfully...");

        }

        catch(IOException e){

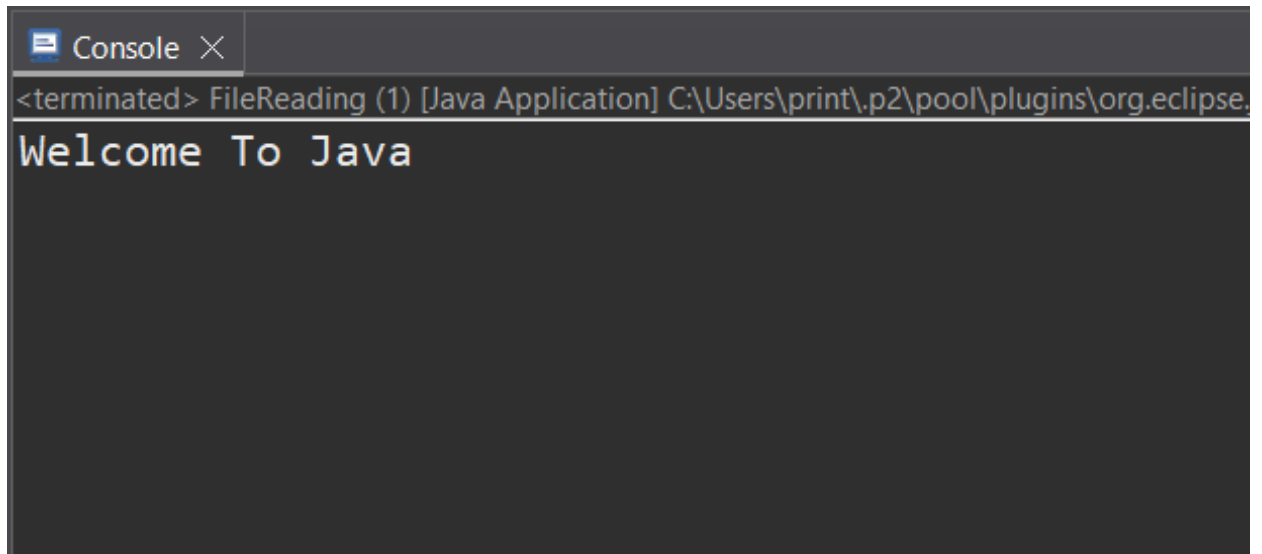
            e.printStackTrace();

        }

    }

}
```

OUTPUT



The screenshot shows a console window titled "Console" with a close button. The output text is as follows:

```
<terminated> FileReading (1) [Java Application] C:\Users\print\.p2\pool\plugins\org.eclipse.  
Welcome To Java
```

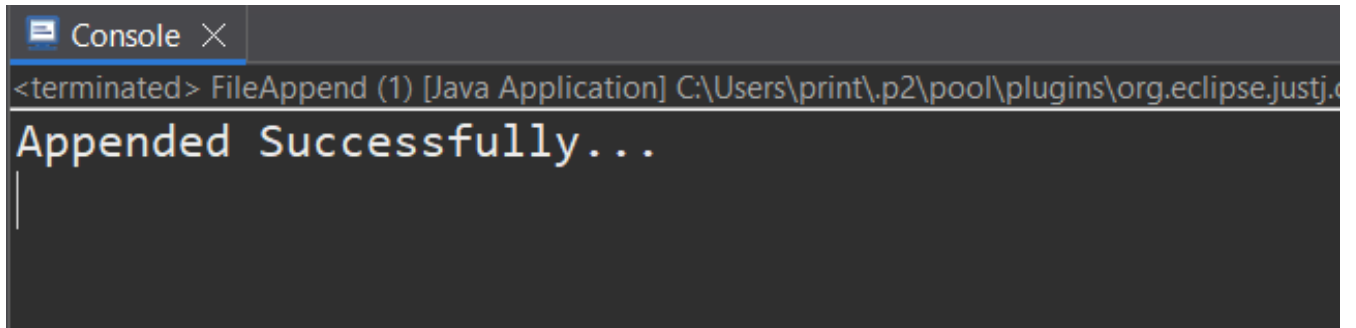
PROGRAM – 17

AIM : Write a Java program to read and display the contents of a text file.

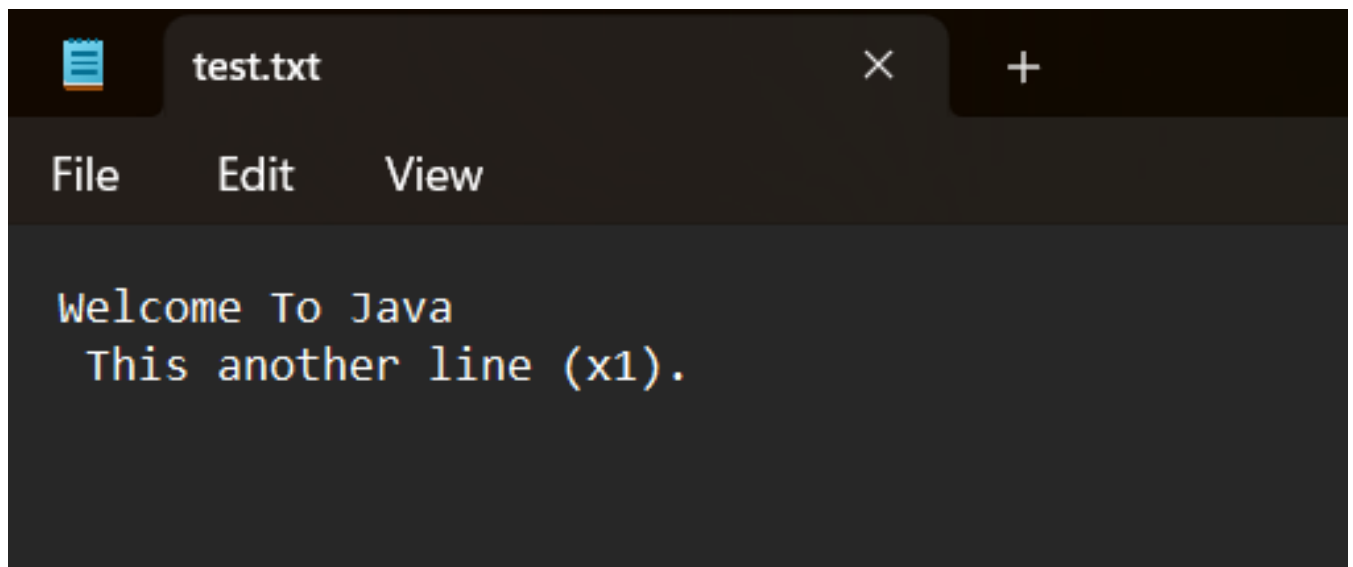
SOURCE CODE:

```
package labprgms;
import java.io.*;
import java.util.Scanner;
public class FileReading {
    public static void main(String[] args) {
        // File Reading
        File myFile = new File("test.txt");
        try {
            Scanner sc = new Scanner(myFile);
            while(sc.hasNextLine())
            {
                System.out.println(sc.nextLine());
            }
        }
        catch (FileNotFoundException e){
            e.printStackTrace();
        }
    }
}
```

OUTPUT



The screenshot shows the Eclipse IDE's Console window. The title bar reads 'Console' with a close button. The text in the console is: `<terminated> FileAppend (1) [Java Application] C:\Users\print\.p2\pool\plugins\org.eclipse.justj.c` followed by a new line and the text `Appended Successfully...` in a larger font.



The screenshot shows the Eclipse IDE's editor window. The title bar reads 'test.txt' with a close button and a plus sign. The menu bar shows 'File', 'Edit', and 'View'. The text in the editor is: `Welcome To Java` followed by a new line and `This another line (x1).`

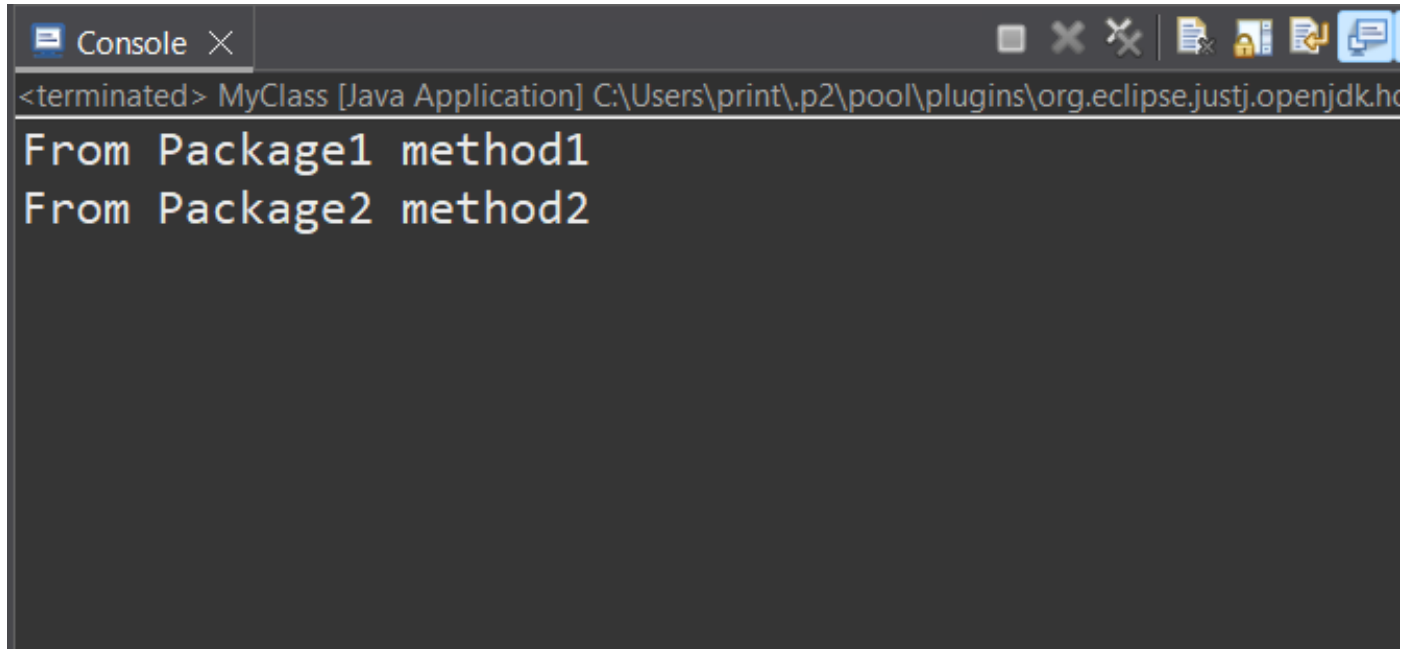
PROGRAM – 18

AIM : Write a Java program to append data to an existing file without overwriting its contents.

SOURCE CODE:

```
package labprgms;
import java.io.*;
public class FileAppend {
    public static void main(String[] args) {
        try {
            // append() --> add text to the end of the file
            FileWriter writer = new FileWriter("test.txt",true);
            writer.append("\n This another line (x1).");
            writer.close();
            System.out.println("Appended Successfully...");
        }
        catch(IOException e){
            e.printStackTrace();
        }
    }
}
```

OUTPUT



The screenshot shows a console window titled "Console" with a close button. The window contains the following text:

```
<terminated> MyClass [Java Application] C:\Users\print\.p2\pool\plugins\org.eclipse.justj.openjdk.h  
From Package1 method1  
From Package2 method2
```

PROGRAM – 19

AIM : Write a Java program to create and use a user-defined package containing classes and methods.

SOURCE CODE:

Create two packages: “package1” & “package2” and a java file inside it.

ClassA.java

```
package package1;

public class ClassA {
    public void method1() {
        System.out.println("From Package1 method1");
    }
}
```

ClassB.java

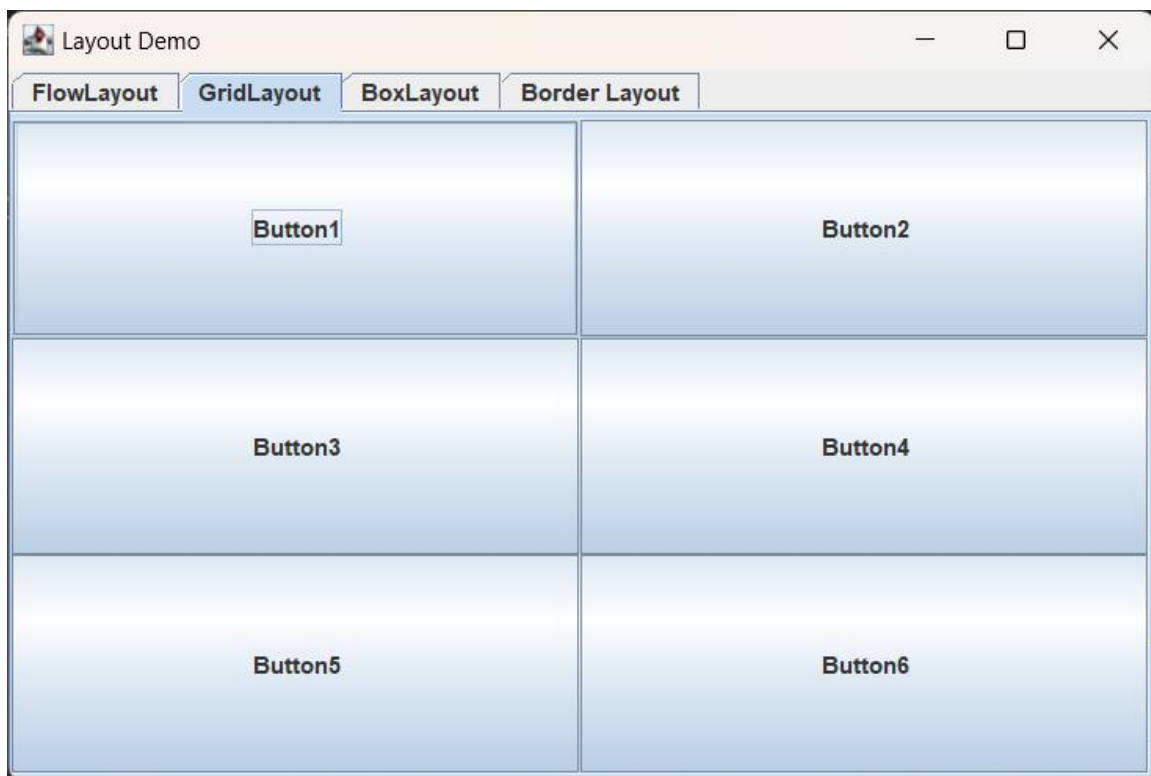
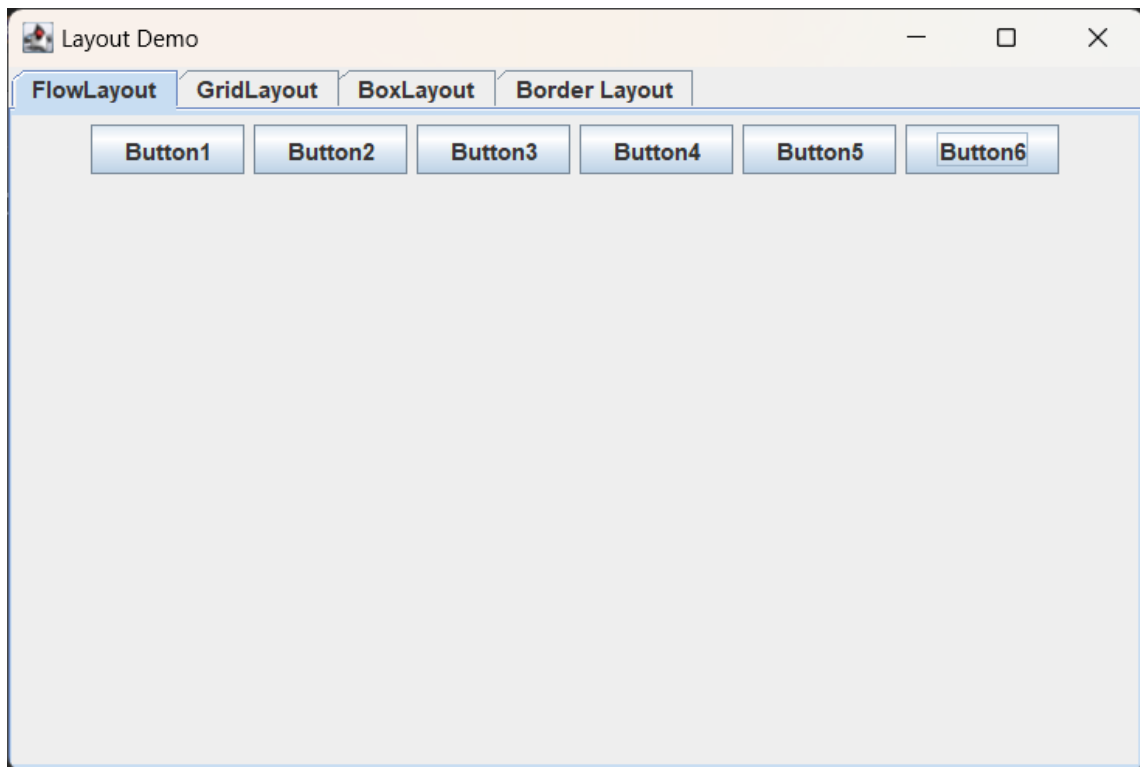
```
package package2;

public class ClassB {
    public void method2() {
        System.out.println("From Package2 method2");
    }
    private void method3() {
        System.out.println("From Package2 method3");
    }
    protected void method4() {
        System.out.println("From Package2 method4");
    }
}
```


MyClass.java

```
package labprgms;
import package1.ClassA;
import package2.ClassB;
import java.util.Scanner;
public class MyClass {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ClassA cla = new ClassA();
        cla.method1();
        ClassB clb = new ClassB();
        clb.method2();
    }
}
```

OUTPUT



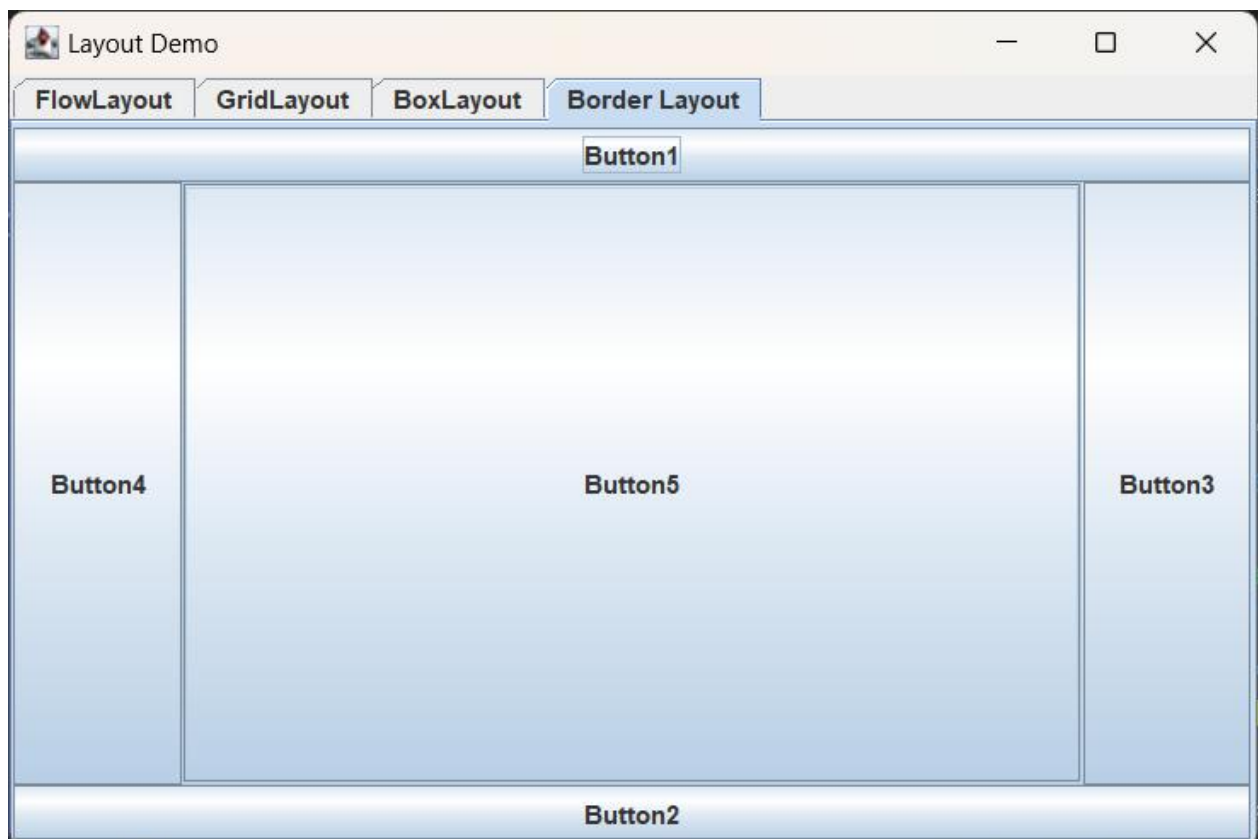
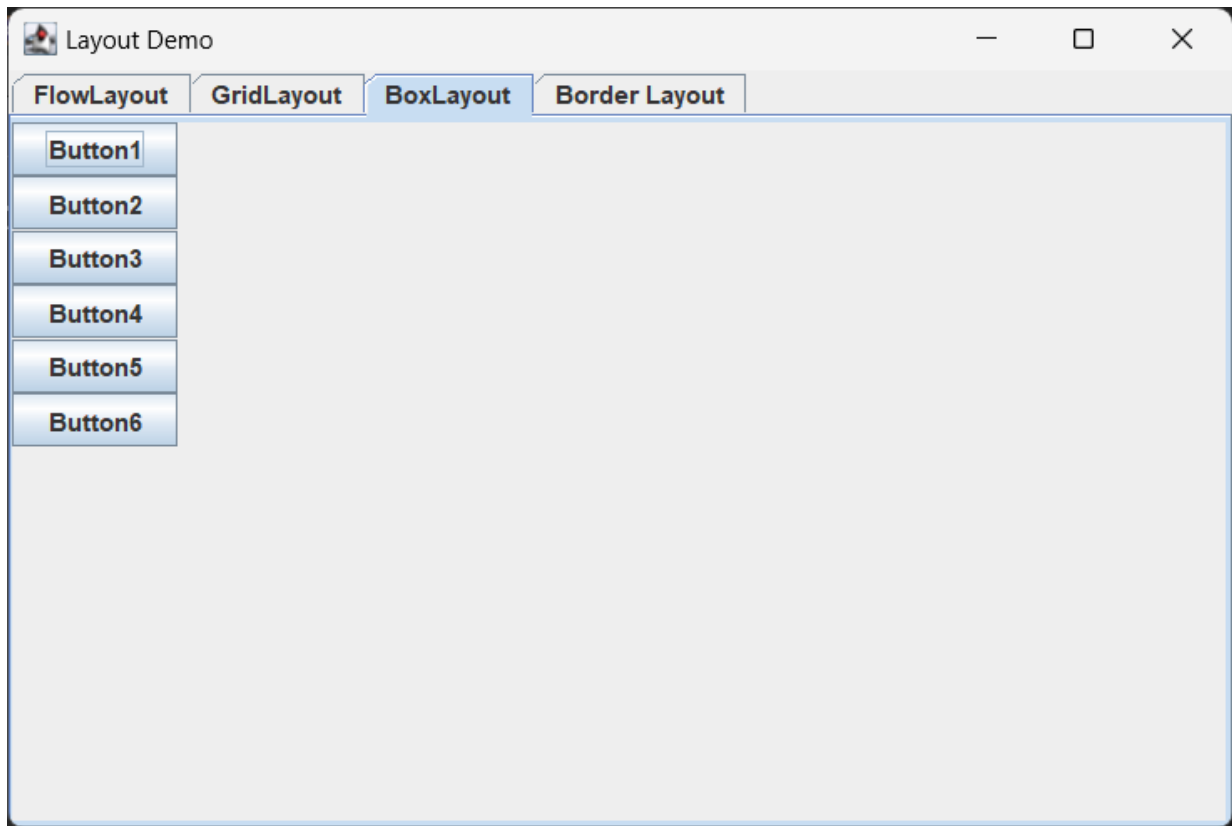
PROGRAM – 20

AIM : Write a Java program to demonstrate the use of layout managers for arranging GUI components.

SOURCE CODE:

```
package labprgms;
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
public class LayoutManagersDemo {
    JFrame f;
    public LayoutManagersDemo() {
        f=new JFrame("Layout Demo");
        f.setSize(600,400);
        JTabbedPane tab = new JTabbedPane();
        JPanel panel1 = createFlowLayout();
        tab.addTab("FlowLayout", panel1);
        JPanel panel2 = createGridLayout();
        tab.addTab("GridLayout", panel2);
        JPanel panel3 = createBoxLayout();
        tab.addTab("BoxLayout", panel3);
        JPanel panel4 = createBorderLayout();
        tab.addTab("Border Layout", panel4);
        f.add(tab);
        f.setVisible(true);
    }
    private JPanel createBorderLayout() {
        JPanel myPanel = new JPanel();
        myPanel.setLayout(new BorderLayout());
        myPanel.add(new JButton("Button1"), BorderLayout.NORTH);
        myPanel.add(new JButton("Button2"), BorderLayout.SOUTH);
        myPanel.add(new JButton("Button3"), BorderLayout.EAST);
        myPanel.add(new JButton("Button4"), BorderLayout.WEST);
        myPanel.add(new JButton("Button5"), BorderLayout.CENTER);
    }
}
```

OUTPUT



```

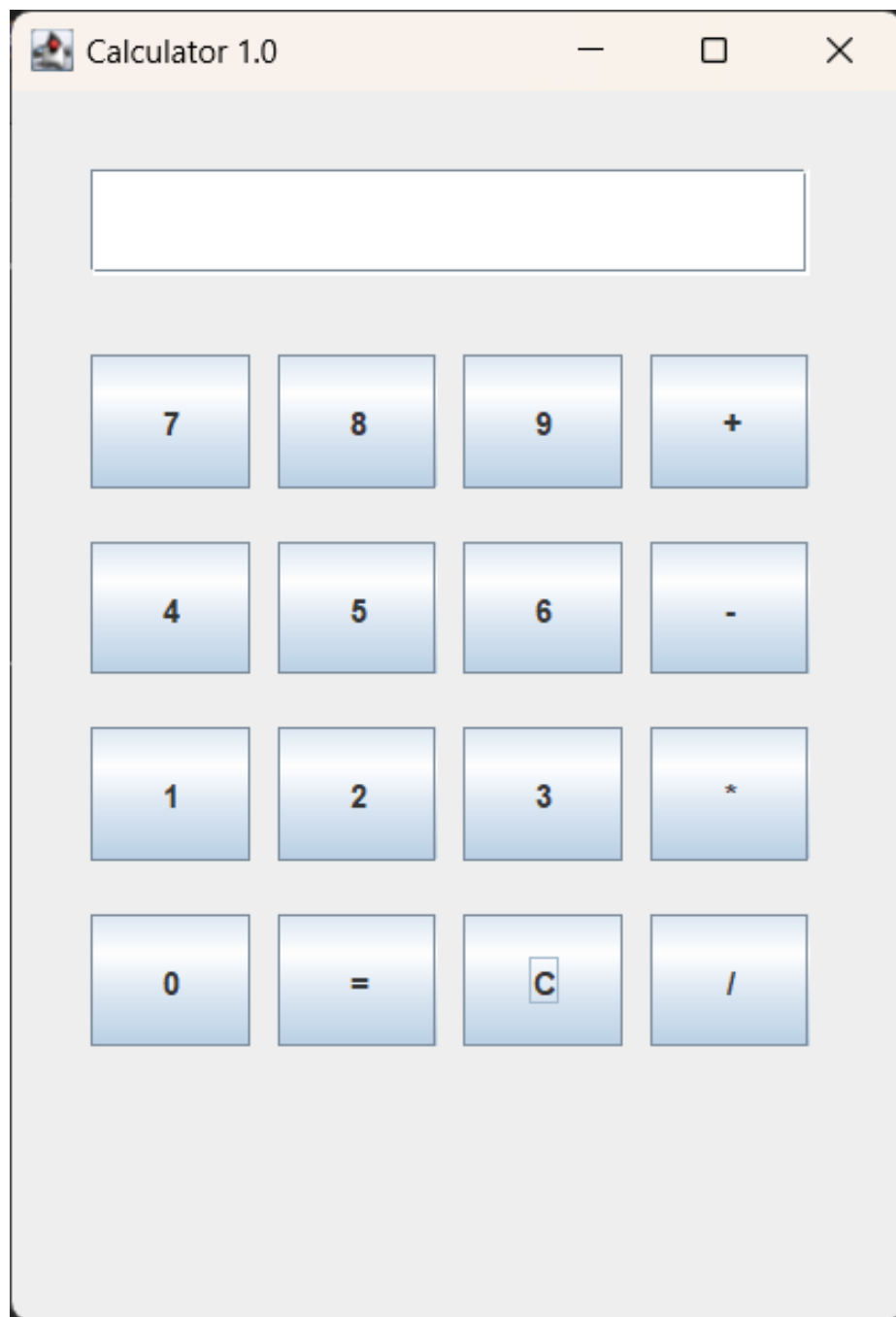
        return myPanel;
    }
    private JPanel createBoxLayout() {
        JPanel myPanel = new JPanel();
        myPanel.setLayout(new BoxLayout(myPanel, BoxLayout.Y_AXIS));
        myPanel.add(new JButton("Button1"));
        myPanel.add(new JButton("Button2"));
        myPanel.add(new JButton("Button3"));
        myPanel.add(new JButton("Button4"));
        myPanel.add(new JButton("Button5"));
        myPanel.add(new JButton("Button6"));
        return myPanel;
    }
    private JPanel createGridLayout() {
        JPanel myPanel = new JPanel();
        myPanel.setLayout(new GridLayout(3,2));
        myPanel.add(new JButton("Button1"));
        myPanel.add(new JButton("Button2"));
        myPanel.add(new JButton("Button3"));
        myPanel.add(new JButton("Button4"));
        myPanel.add(new JButton("Button5"));
        myPanel.add(new JButton("Button6"));
        return myPanel;
    }
    private JPanel createFlowLayout() {
        JPanel myPanel = new JPanel();
        myPanel.setLayout(new FlowLayout());
        myPanel.add(new JButton("Button1"));
        myPanel.add(new JButton("Button2"));
        myPanel.add(new JButton("Button3"));
        myPanel.add(new JButton("Button4"));
        myPanel.add(new JButton("Button5"));
        myPanel.add(new JButton("Button6"));
        return myPanel;
    }

```



```
    }  
    public static void main(String[] args) {  
        new LayoutManagersDemo();  
    }  
}
```

OUTPUT



PROGRAM – 21

AIM : Write a Java program to develop a simple GUI calculator that performs basic arithmetic operations.

SOURCE CODE:

```
package labprgms;
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;

public class GuiCalculator implements ActionListener {
    JFrame f;
    JTextField tf;
    JButton b0, b1, b2, b3, b4, b5, b6, b7, b8, b9;
    JButton add, sub, mul, div, equal, clear;
    double num1, num2, result;
    String operator;

    public GuiCalculator() {
        f = new JFrame("Calculator 1.0");
        f.setSize(350, 500);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // Text Field
        tf = new JTextField();
        tf.setBounds(30, 30, 270, 40);
        tf.setFont(new Font("Arial", Font.BOLD, 20));
        f.add(tf);

        // Number Buttons
        b7 = new JButton("7");
        b7.setBounds(30, 100, 60, 50);
        b8 = new JButton("8");
        b8.setBounds(100, 100, 60, 50);
        b9 = new JButton("9");
        b9.setBounds(170, 100, 60, 50);
        add = new JButton("+");
```



```

add.setBounds(240, 100, 60, 50);
b4 = new JButton("4");
b4.setBounds(30, 170, 60, 50);
b5 = new JButton("5");
b5.setBounds(100, 170, 60, 50);
b6 = new JButton("6");
b6.setBounds(170, 170, 60, 50);
sub = new JButton("-");
sub.setBounds(240, 170, 60, 50);
b1 = new JButton("1");
b1.setBounds(30, 240, 60, 50);
b2 = new JButton("2");
b2.setBounds(100, 240, 60, 50);
b3 = new JButton("3");
b3.setBounds(170, 240, 60, 50);
mul = new JButton("*");
mul.setBounds(240, 240, 60, 50);
b0 = new JButton("0");
b0.setBounds(30, 310, 60, 50);
equal = new JButton("=");
equal.setBounds(100, 310, 60, 50);
clear = new JButton("C");
clear.setBounds(170, 310, 60, 50);
div = new JButton("/");
div.setBounds(240, 310, 60, 50);
// Add Buttons
f.add(b0);
f.add(b1);
f.add(b2);
f.add(b3);
f.add(b4);
f.add(b5);
f.add(b6);
f.add(b7);

```



```

f.add(b8);
f.add(b9);
f.add(add);
f.add(sub);
f.add(mul);
f.add(div);
f.add(equal);
f.add(clear);
// Action Listeners
b0.addActionListener(this);
b1.addActionListener(this);
b2.addActionListener(this);
b3.addActionListener(this);
b4.addActionListener(this);
b5.addActionListener(this);
b6.addActionListener(this);
b7.addActionListener(this);
b8.addActionListener(this);
b9.addActionListener(this);
add.addActionListener(this);
sub.addActionListener(this);
mul.addActionListener(this);
div.addActionListener(this);
equal.addActionListener(this);
clear.addActionListener(this);
f.setVisible(true);
}

public static void main(String[] args) {
    new GuiCalculator();
}

```


@Override

```
public void actionPerformed(ActionEvent e) {  
    // Number Buttons  
    if (e.getSource() == b0)  
        tf.setText(tf.getText() + "0");  
    if (e.getSource() == b1)  
        tf.setText(tf.getText() + "1");  
    if (e.getSource() == b2)  
        tf.setText(tf.getText() + "2");  
    if (e.getSource() == b3)  
        tf.setText(tf.getText() + "3");  
    if (e.getSource() == b4)  
        tf.setText(tf.getText() + "4");  
    if (e.getSource() == b5)  
        tf.setText(tf.getText() + "5");  
    if (e.getSource() == b6)  
        tf.setText(tf.getText() + "6");  
    if (e.getSource() == b7)  
        tf.setText(tf.getText() + "7");  
    if (e.getSource() == b8)  
        tf.setText(tf.getText() + "8");  
    if (e.getSource() == b9)  
        tf.setText(tf.getText() + "9");  
    // Operators  
    if (e.getSource() == add) {  
        num1 = Double.parseDouble(tf.getText());  
        operator = "+";  
        tf.setText("");  
    }  
    if (e.getSource() == sub) {  
        num1 = Double.parseDouble(tf.getText());  
        operator = "-";  
        tf.setText("");  
    }  
}
```

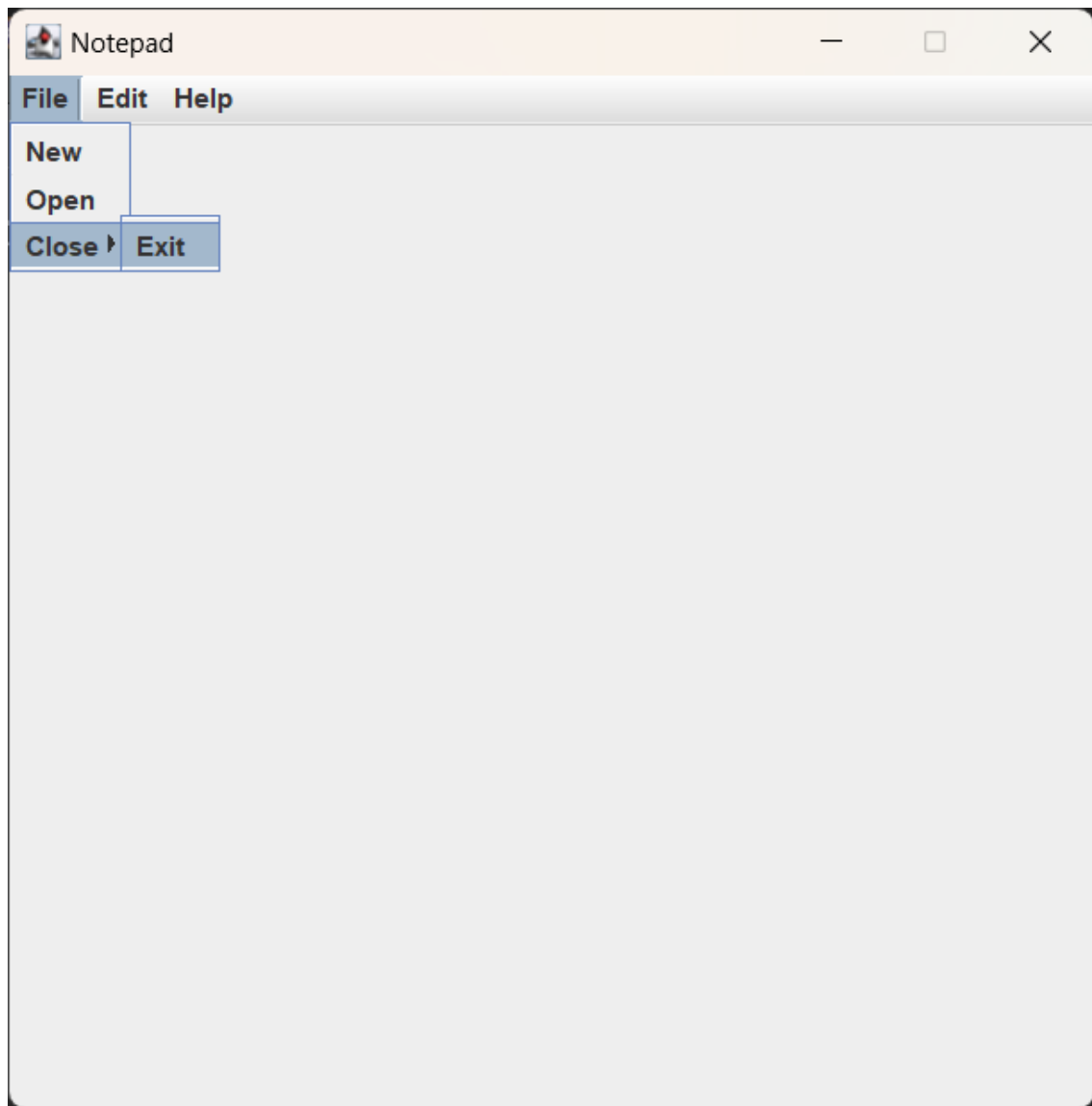


```

if (e.getSource() == mul) {
    num1 = Double.parseDouble(tf.getText());
    operator = "*";
    tf.setText("");
}
if (e.getSource() == div) {
    num1 = Double.parseDouble(tf.getText());
    operator = "/";
    tf.setText("");
}
// Equal Button
if (e.getSource() == equal) {
    num2 = Double.parseDouble(tf.getText());
    if (operator.equals("+"))
        result = num1 + num2;
    if (operator.equals("-"))
        result = num1 - num2;
    if (operator.equals("*"))
        result = num1 * num2;
    if (operator.equals("/"))
        result = num1 / num2;
    tf.setText("" + result);
}
// Clear Button
if (e.getSource() == clear) {
    tf.setText("");
}
}
}

```

OUTPUT



PROGRAM – 22

AIM : Write a Java program to create a menu-driven GUI application using menu bars, menus, and menu items.

SOURCE CODE:

```
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.*;

public class MenuCreation implements ActionListener{
    JFrame f;
    JMenuBar mb;
    JMenu fileMenu,editMenu,helpMenu, closeMenu;
    JMenuItem newFile, openFile, exitFile;
    public MenuCreation() {

        f = new JFrame("Notepad");
        f.setSize(500,500);
        f.setResizable(false);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        mb = new JMenuBar();
        f.setJMenuBar(mb);
        fileMenu = new JMenu("File");
        mb.add(fileMenu);
        editMenu = new JMenu("Edit");
        mb.add(editMenu);
        helpMenu = new JMenu("Help");
        mb.add(helpMenu);
        newFile = new JMenuItem("New");
        fileMenu.add(newFile);
        openFile = new JMenuItem("Open");
        fileMenu.add(openFile);
```



```

closeMenu = new JMenu("Close");
fileMenu.add(closeMenu);

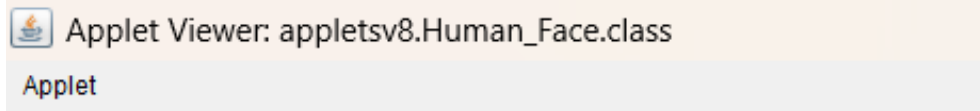
exitFile = new JMenuItem("Exit");
closeMenu.add(exitFile);
exitFile.addActionListener(this);
f.setVisible(true);
}

public static void main(String[] args) {
    new MenuCreation();
}

@Override
public void actionPerformed(ActionEvent e) {
    if(e.getSource()==exitFile) {
        System.exit(0);
    }
}
}

```

OUTPUT



PROGRAM – 23

AIM : Write a Java applet program to draw a human face.

SOURCE CODE:

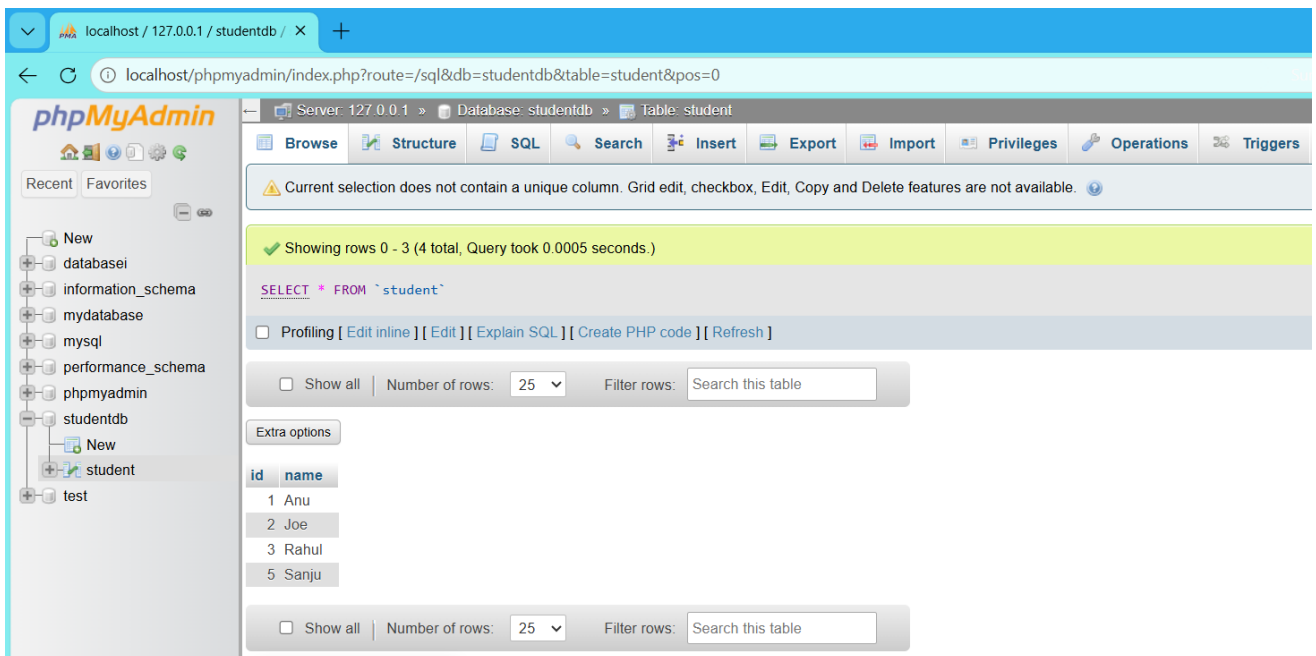
```
package appletsv8;
import java.awt.*;
import java.applet.*;
import javax.swing.JApplet;
public class Human_Face extends Applet
{
    //Initialize the applet
    public void init()
    {
        setBackground(Color.white);
    }
    //Draw the human face
    public void paint(Graphics g)
    {
        //Change color to cream
        Color clr=new Color(255,179,86);
        g.setColor(clr);
        //Draw and fill the face
        g.drawOval(100,100,250,300);
        g.fillOval(100,100,250,300);
        //Change color to black
        g.setColor(Color.black);
        //Draw the left eye
        g.drawOval(160,185,40,25);
        g.fillOval(160,185,40,25);
        //Draw the right eye
        g.drawOval(250,185,40,25);
        g.fillOval(250,185,40,25);
    }
}
```



```
//Draw the Left Eyebrow
g.drawArc(160,170,35,10,0,180);
//Draw the Right Eyebrow
g.drawArc(250,170,35,10,0,180);
//Draw the Nose
g.drawLine(210,265,210,275);
g.drawLine(240,265,240,275);
g.drawArc(210,275,30,10,0,-180);
//Draw the smile
g.drawArc(175,300,100,50,0,-180);
}
}
/*
<applet code = Human_Face.class width=500 height=500>
</applet>
*/
```

OUTPUT

```
Console X
<terminated> DatabaseExample [Java Application] C:\Users\print\p2\pool\plugins\org.eclipse.justj.openjdk.hotsp
Connected
Id: 1 , Name: Anu
Id: 2 , Name: Binu
Id: 3 , Name: Rahul
Id: 4 , Name: Manu
Inserted
Deleted
Updated Succ...
```



Database

PROGRAM – 24

AIM : Write a Java program to establish connectivity with a MySQL database using JDBC and perform basic database operations.

SOURCE CODE:

```
package labprgms;
import java.sql.*;
public class DatabaseExample {
    public static void main(String[] args) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection conn =
DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb?", "root", "");
            System.out.println("Connected");
            // Print all from db
            String query = "Select * from student";
            PreparedStatement pstmt = conn.prepareStatement(query);
            ResultSet rs =pstmt.executeQuery();
            while(rs.next()) {
                int id = rs.getInt("id");
                String name = rs.getString("name");
                System.out.println("Id: "+id +", " + "Name: "+name);
            }

            //insert
            String insert = "INSERT INTO student (id,name) VALUES(?,?)";
            pstmt = conn.prepareStatement(insert);
            pstmt.setInt(1, 6);
            pstmt.setString(2, "Biju");
            pstmt.executeUpdate();
            System.out.println("Inserted");
```



```

// delete
String delete = "DELETE FROM STUDENT WHERE id = 2";
pstmt = conn.prepareStatement(delete);
pstmt.executeUpdate();
System.out.println("Deleted");

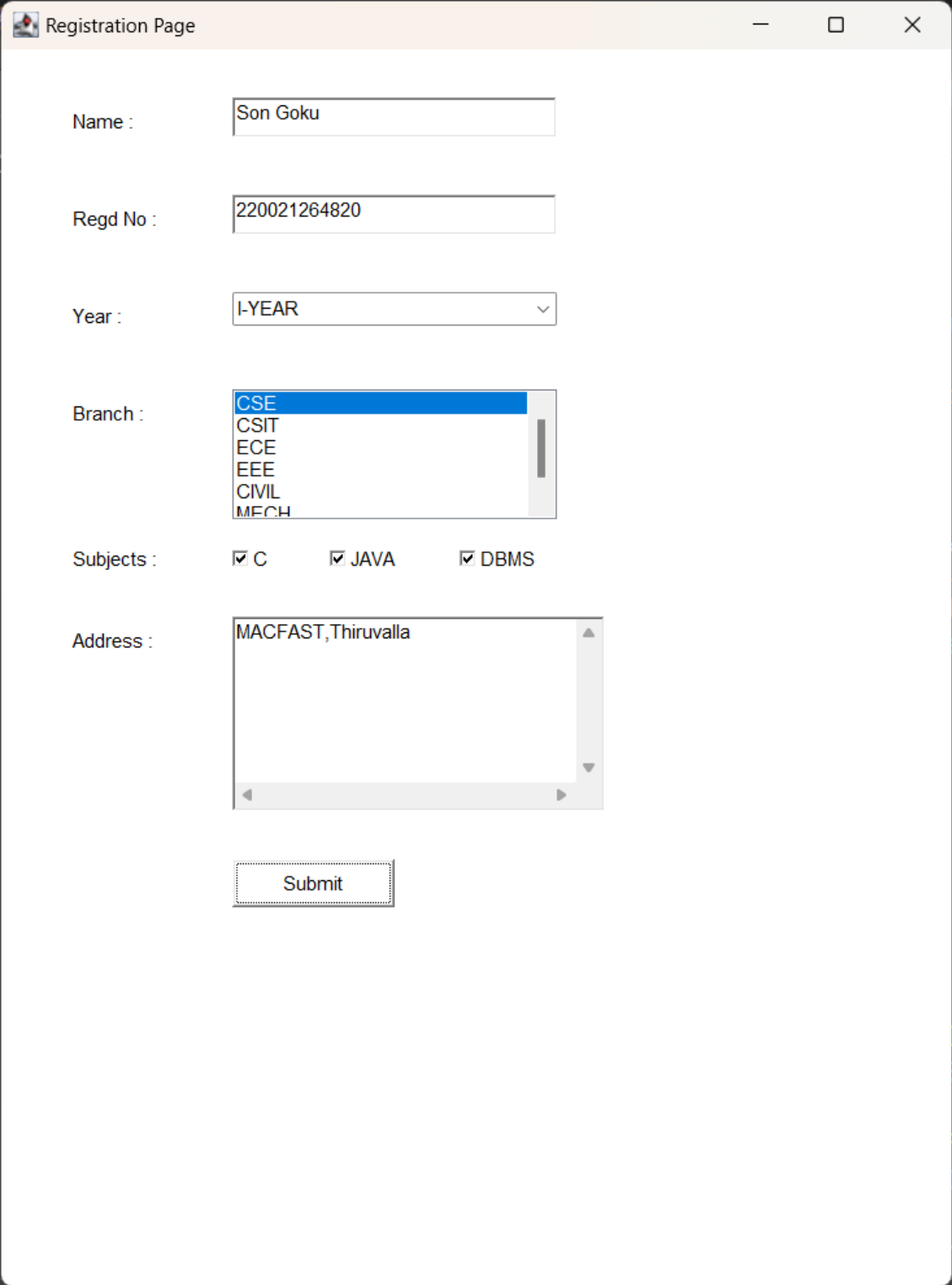
// update
String update = "UPDATE STUDENT SET name = 'Joe' WHERE id = 3";
pstmt = conn.prepareStatement(update);
pstmt.executeUpdate();
System.out.println("Updated Succ...");
} catch (Exception e) {
    System.out.println("Not able to connect");
}

}

}

```

OUTPUT



The screenshot shows a web browser window with the title "Registration Page". The page contains several form fields and a submit button. The fields are labeled "Name :", "Regd No :", "Year :", "Branch :", "Subjects :", and "Address :". The "Name" field contains "Son Goku". The "Regd No" field contains "220021264820". The "Year" field is a dropdown menu showing "I-YEAR". The "Branch" field is a list box with "CSE" selected. The "Subjects" field has three checkboxes: "C", "JAVA", and "DBMS", all of which are checked. The "Address" field is a text area containing "MACFAST,Thiruvalla". A "Submit" button is located at the bottom of the form.

Registration Page

Name : Son Goku

Regd No : 220021264820

Year : I-YEAR

Branch : CSE
CSIT
ECE
EEE
CIVIL
MECH

Subjects : ☒ C ☒ JAVA ☒ DBMS

Address : MACFAST,Thiruvalla

Submit

PROGRAM – 25

AIM : Write a Java program using AWT components to design a student registration form.

SOURCE CODE:

```
package labprgms;
import java.awt.*;
class RegistrationExample
{
    public static void main(String args[])
    {
        Frame f = new Frame();
        f.setSize(600, 800);
        f.setLayout(null);
        f.setVisible(true);
        f.setTitle("Registration Page");
        Label l1, l2, l3, l4, l5, l6;
        TextField t1, t2;
        Choice ch;
        List ls;
        Checkbox cb1, cb2, cb3;
        TextArea ta;
        Button b;
        // Name
        l1 = new Label("Name :");
        l1.setBounds(50, 60, 100, 30);
        t1 = new TextField();
        t1.setBounds(150, 60, 200, 25);
        // Registration Number
        l2 = new Label("Regd No :");
        l2.setBounds(50, 120, 100, 30);
        t2 = new TextField();
        t2.setBounds(150, 120, 200, 25);
```



```

// Year
l3 = new Label("Year :");
l3.setBounds(50, 180, 100, 30);
ch = new Choice();
ch.setBounds(150, 180, 200, 30);
ch.add("I-YEAR");
ch.add("II-YEAR");
ch.add("III-YEAR");
ch.add("IV-YEAR");
// Branch
l4 = new Label("Branch :");
l4.setBounds(50, 240, 100, 30);
ls = new List(4);
ls.setBounds(150, 240, 200, 80);
ls.add("CSE");
ls.add("CSIT");
ls.add("ECE");
ls.add("EEE");
ls.add("CIVIL");
ls.add("MECH");
// Subjects
l5 = new Label("Subjects :");
l5.setBounds(50, 330, 100, 30);
cb1 = new Checkbox("C");
cb1.setBounds(150, 330, 50, 30);
cb2 = new Checkbox("JAVA");
cb2.setBounds(210, 330, 70, 30);
cb3 = new Checkbox("DBMS");
cb3.setBounds(290, 330, 80, 30);
// Address
l6 = new Label("Address :");
l6.setBounds(50, 380, 100, 30);
ta = new TextArea();
ta.setBounds(150, 380, 230, 120);

```



```
// Submit Button
b = new Button("Submit");
b.setBounds(150, 530, 100, 30);
// Adding components to frame
f.add(l1);
f.add(l2);
f.add(l3);
f.add(l4);
f.add(l5);
f.add(l6);
f.add(t1);
f.add(t2);
f.add(ch);
f.add(ls);
f.add(cb1);
f.add(cb2);
f.add(cb3);
f.add(ta);
f.add(b);
}
}
```

OUTPUT

```
C:\Windows\System32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\print\Downloads>javac MyServer.java

C:\Users\print\Downloads>java MyServer
Waiting for client
|
```

```
C:\Windows\System32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\print\Downloads>javac MyClient.java

C:\Users\print\Downloads>java MyClient
```

PROGRAM – 26

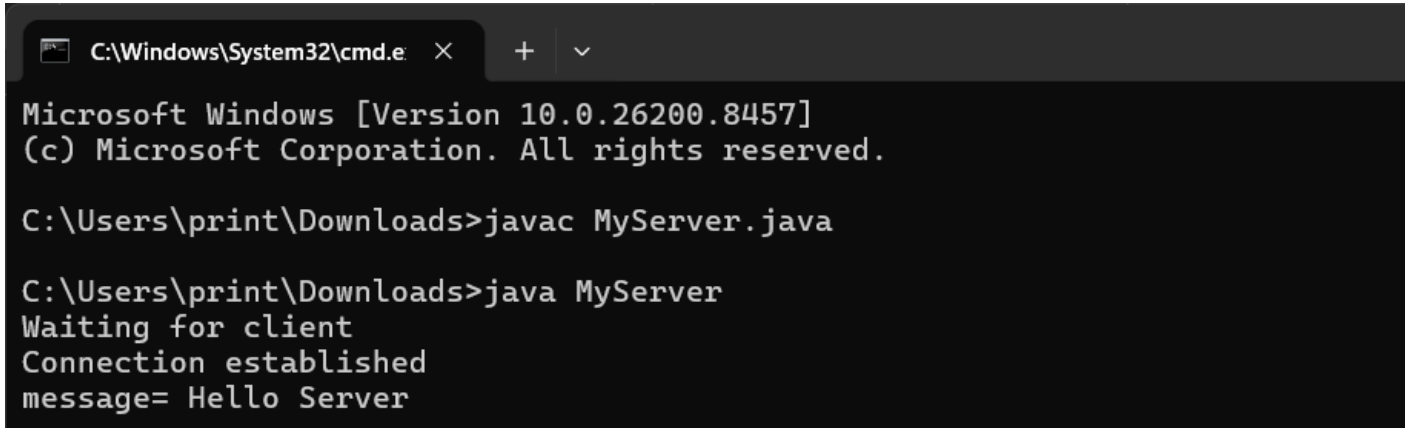
AIM : Write a Java program to demonstrate client-server communication using socket programming.

SOURCE CODE:

MyServer.java

```
package labprgms;
import java.io.*;
import java.net.*;
public class MyServer {
    public static void main(String[] args){
        try{
            ServerSocket ss=new ServerSocket(6666);
            System.out.println("Waiting for client");
            Socket s=ss.accept();//establishes connection
            System.out.println("Connection established");
            DataInputStream dis=new DataInputStream(s.getInputStream());
            String str=(String)dis.readUTF();
            System.out.println("message= "+str);
            ss.close();
        }
        catch(Exception e){
            System.out.println(e);
        }
    }
}
```

OUTPUT



```
C:\Windows\System32\cmd.e  ×  +  ∨  
Microsoft Windows [Version 10.0.26200.8457]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\print\Downloads>javac MyServer.java  
  
C:\Users\print\Downloads>java MyServer  
Waiting for client  
Connection established  
message= Hello Server
```

MyClient.java

```
package labprgms;

import java.io.*;
import java.net.*;

public class MyClient {
    public static void main(String[] args) {
        try{
            Socket s=new Socket("localhost",6666);
            DataOutputStream dout=new DataOutputStream(s.getOutputStream());
            dout.writeUTF("Hello Server");
            dout.flush();
            dout.close();
            s.close();
        }catch(Exception e){
            System.out.println(e);
        }
    }
}
```